

Assembly Without Industry: The Gap Between Policy and Production in Pakistan's Electric Vehicle and Battery Sector

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Executive Summary

Pakistan is attempting to convert a rapidly growing demand for imported batteries and electric vehicles into a domestic manufacturing industry. This report asks whether the industrial and policy base exists to support that conversion, and what a Chinese investor can defensibly commit to today. It examines the six policy instruments governing the sector and the four draft incentive lines proposed for battery manufacturing, the market data, the supply chain and mineral base, six named Chinese engagements, and the competing offers of Bangladesh, Vietnam, Indonesia and Thailand.

The demand is real. Lithium-ion storage imports reached 4.6 GWh in 2025, a 220% increase year on year, and that single year accounted for about 60% of everything imported since 2018.¹ Distributed solar capacity is estimated at about 38 GW, 27 GW of it installed in the two years to June 2025.² At current petrol prices and reported public charging prices, running an electric car on public chargers costs roughly 38% to 83% less than running a petrol equivalent, depending on the operator and charger type, and about 92% less when it is charged from rooftop solar. A country importing this much energy hardware, facing that cost differential, has an obvious reason to want the factories rather than the shipments.

The industrial base does not yet exist. Pakistan has no commercial cell manufacturing. Its first dedicated lithium battery plant, at Korangi in Karachi, runs at roughly 4 MW initial capacity and about 2,000 units a month, which is a useful start but not an industry. Every other facility described publicly as battery manufacturing is assembling imported cells into packs, and the same pattern holds on the vehicle side. This report distinguishes assembly from manufacturing, and by that distinction essentially all current Pakistani activity falls on the assembly side.

The policy framework intended to close that gap is largely unfinished, and this is the finding with the greatest bearing on investment decisions. Of the six instruments governing the sector, three are notified and enforceable, two rest on agreement in principle, and one exists only in draft. None of the three enforceable instruments governs battery manufacturing, and all four incentive lines proposed for battery manufacturing exist only in an investment pitchbook. An investor building a financial model on the pitchbook figures is underwriting an instrument that does not yet exist.

Where Pakistan does have a notified policy, its record is the best available evidence on execution risk. The New Energy Vehicle (NEV) Policy 2025-30 was approved by the Economic Coordination Committee in August 2025 and is financed through a statutory levy of 1% to 3% on new petrol and diesel vehicles. The pilot of the subsidy programme beneath it targeted 41,000 vehicles and delivered 5,409, or 13%, before being redesigned; measured against the full year target of 119,170 vehicles, delivery was about 4.5%. The

bottleneck was neither demand nor capacity but a payment design that made buyers pay in full and reclaim afterwards. The government reversed that flow within roughly nine months.

Chinese engagement is real but concentrated at the assembly end. The largest funded, financially closed project is a completely knocked down (CKD) plant at Gharo in Sindh, at approximately USD 150 million for 25,000 units a year, with BYD supplying technology and branding under licence rather than taking equity; its launch has already slipped once. Two Chery-linked operations assemble plug-in hybrids, one of them at a USD 100 million plant in Faisalabad. On the battery side, Treet Battery has a memorandum with a company in which Contemporary Amperex Technology Company (CATL) holds a strategic investment, which is a commercial relationship rather than a commitment to produce cells. The mineral case for the chosen lithium iron phosphate chemistry is sound in its geology and unproven in its metallurgy: Pakistan has the ore but has not shown it can produce the material.

Set against Bangladesh, Vietnam, Indonesia and Thailand, which compete for the same capital, Pakistan offers a large underpenetrated market and the thinnest competitive field, but the least legal certainty, average industrial electricity prices well above those in India and China, and a recent record of import and repatriation controls. The investable opportunities today are therefore not cell manufacturing but the layers beneath it: pack and module assembly, battery management systems, thermal systems, testing and certification, recycling, charging infrastructure, and battery-grade processing of the iron and phosphate base. Investment should be phased against notified rather than announced policy, with renegotiation triggers built into anything signed before the pending instruments are gazetted. Where a choice exists, a tariff-based incentive is worth more than a disbursement-based one even at a lower headline value, because it operates automatically at the border while the disbursement route depends on the administrative chain that delivered 13% of its pilot target.

1 Research Question and Objectives

Pakistan's electric vehicle and battery sector generates a volume of announcement disproportionate to the volume of verifiable industrial activity. Headline targets, localisation percentages and memoranda of understanding circulate widely and are frequently reproduced as though they were settled policy or committed capital. The distance between what has been announced and what can be relied upon is, for an investor, the whole of the question.

To what extent does Pakistan's policy framework, market evidence, supply chain and mineral base support a Chinese investment in electric vehicle and battery manufacturing, as distinct from assembly, at the date of this report?

Four objectives follow from that question, and the structure of this report follows the objectives.

1. To establish, instrument by instrument, which elements of Pakistan's electric vehicle and battery policy framework are legally binding and which exist only as intention, and to state the evidentiary basis for each classification.
2. To test the headline market claims circulating about Pakistani electric vehicle demand, pricing and running costs against their underlying data, and to identify where the published record is too fragmented to support a conclusion.
3. To apply a consistent classification test separating assembly from manufacturing across the current landscape of Pakistani battery and vehicle projects, including the named Chinese engagements, and to assess the mineral base against battery-grade rather than geological criteria.
4. To situate Pakistan against Bangladesh, Vietnam, Indonesia and Thailand, which compete for the same capital, and to specify the observable developments that would resolve the questions this report must leave open.

2 Methodology

This report is a documentary and data analysis of the public record on Pakistan's electric vehicle and battery sector, conducted from Islamabad, with evidence reviewed to 24 September 2026. It applies two classification frameworks consistently across that record: one separating policy instruments by legal standing, and one separating industrial activity by position on the battery value chain. Neither framework is novel. What the report contributes is their systematic application to a body of evidence that is usually discussed instrument by instrument and project by project, without a common standard against which claims can be compared.

The design is deliberately not predictive. The report does not forecast electric vehicle penetration, battery demand or investment volumes, because the published Pakistani dataset does not support forecasting at a useful confidence level, for reasons set out under limitations below. It instead establishes what is currently true, what is currently claimed, and what would have to be observed for a claim to become true.

2.1 Data Sources and Selection

Sources were selected against a defined source quality standard, which admits official statistical agencies, regulators, ministries, multilateral bodies and established research institutions as primary support, and admits specialist financial press only where an official series is reported to a regulator but not separately published.

Government and regulatory sources include the Ministry of Industries and Production's New Energy Vehicles Policy and the expired Auto Industry Development and Export Policy, the Press Information Department, the National Electric Power Regulatory Authority's charging tariff decision, the Federal Board of Revenue's levy notification, the Engineering Development Board's investment pitchbook for battery manufacturing, and statements by the National Energy Efficiency and Conservation Authority. Geological sources are the Geological Survey of Pakistan, its journal *Geologica*, the Punjab Mineral Company and the United States Geological Survey. Corporate and financial sources include the Pakistan Credit Rating Agency's rating report on Hub Power Holdings and company statements by HUBCO and CATL. Comparator sources include Vietnam's Corporate Income Tax Law 67 of 2025, Bangladesh's 2026-27 budget as reported by its state news agency, and multilateral and research sources including the International Monetary Fund, the United States International Trade Commission, the International Energy Agency and the International Institute for Sustainable Development.

Market data derives from Arif Habib Limited's monthly tracking of Pakistan Automotive Manufacturers Association data, as reported by ProPakistani and PakWheels, and from manufacturer ex-factory price lists. Analytical sources include Renewables First, Ember, the Institute for Energy Economics and Financial Analysis (IEEFA) and peer reviewed research on the energy use of battery cell manufacturing. The mineral assessment in section 5.3 relies on survey material at the level of documented occurrence only; no reserve tonnage is relied upon.

Press coverage from Dawn, Business Recorder, the Express Tribune, Profit, ProPakistani, Arab News, the Daily Star, Prothom Alo and regional outlets has been used where primary documentation was unavailable, and is identified as such at the point of use. The reference list is weighted toward the specialist press, and the reason should be stated: much of Pakistan's electric vehicle and battery policy has been made public first, and sometimes only, through ministerial statements and briefings reported by the press, and several of the instruments examined here have no published text. Where a primary document is available

online it is cited in preference to press reporting of that document. Foreign tax law is cited through the published analyses of professional advisory firms where the legal text is not available in English. Where two credible sources conflict, both are reported and the reasoning for the figure adopted is stated rather than resolved silently. Where a figure could not be verified against a primary or credible secondary source, it has been excluded rather than reproduced with a qualifier.

2.2 Definitions and Scope

Every policy measure discussed in this report carries a stated legal standing, and three terms are used consistently throughout. Notified means the measure has cleared cabinet, a cabinet committee or the legislature and is in force at the date of writing. In principle means a committee or steering group has agreed its shape but it has not been formally approved. Draft or pitchbook means the figure appears only in a promotional or consultative document and carries no legal force. Where the evidence does not establish which of the three applies, the report says so rather than inferring.

The same discipline governs the distinction between assembly and manufacturing used throughout. A facility that imports finished cells and builds them into modules and packs is classified as assembly. Only a facility producing cells from active materials under local process control is classified as manufacturing. This is not a rhetorical distinction. It determines the capital intensity of a project, the technical capability that must be transferred, the local value actually created, and the policy treatment a project can defensibly claim.

Figures identified as analytical scoring represent this report's assessment of the reviewed evidence rather than measured production shares or official rankings, and are identified as such in their captions. Figures 6, 13 and 15 fall into this category, and the scale anchors and evidence behind every score are set out in Appendix C so that each score can be checked and contested. The scoring in each case is ordinal rather than cardinal: it ranks the inputs against one another on the evidence reviewed and does not claim measured precision at the level of an individual score.

Comparators were selected against a stated rule: each is a developing economy that has adopted an electric vehicle or battery incentive framework since 2023 and is competing for Chinese electric vehicle or battery capital. Bangladesh is included as the closest South Asian peer, with an assembly-led automotive sector and a newly tiered incentive regime. Vietnam is included as an established export manufacturing destination with a newly codified incentive law.

Indonesia is included as the resource-led case, the country most often cited as the template for converting mineral endowment into battery manufacturing. Thailand is included as the country in which Chinese electric vehicle makers have built the most assembly capacity in the region. India is excluded because Chinese investment that confers control or more than 10% beneficial ownership still requires prior government approval, and the fast track introduced in March 2026 does not name electric vehicles or batteries among its sectors;³ it is not an open alternative for the investors this report addresses.

Scope is limited to battery electric and plug-in hybrid vehicles and to lithium-ion battery and energy storage manufacturing. Conventional hybrids appear only where their tax treatment illuminates government preference. Two- and three-wheelers are treated where the subsidy record and local content position bear on the wider argument, but the report is not a survey of that segment. Geographic scope is national, with provincial material introduced where it is material to siting.

2.3 Limitations

Five limitations bear on what this method can establish, and they are stated here rather than left implicit in the findings.

First, the absence of a centralised dataset. Pakistan has no single published series combining monthly electric, hybrid and plug-in hybrid registrations by brand and province. Any national picture must be triangulated across manufacturer disclosures, provincial registrations, customs statistics and company announcements, each capturing a different and partially overlapping slice of the market. The report therefore reports ranges and identifies conflicts rather than presenting reconciled totals that the underlying data cannot support.

Second, unfinished policy. Several of the instruments assessed here are in motion. A classification stated in this report is a classification at the date of writing, and the trigger analysis in section 9 exists precisely so that the classification can be updated against observable events rather than requiring the whole analysis to be redone.

Third, the limits of documentary method on questions of intent and capability. This report can establish what a project has publicly committed to and what it has publicly disclosed. It cannot establish the actual bill of materials of an operating plant, the real domestic content of a vehicle not yet in production, or the technical depth of a technology transfer arrangement whose terms are not

public. Where those questions are decisive, the report says what would have to be verified and by whom, rather than substituting inference.

Fourth, no primary interviews were conducted for this report and no confidential corporate material was reviewed. The analysis rests entirely on the public record. A due diligence exercise with access to project documentation would be able to resolve several of the open items identified here, and the report is written to be useful as a framework for such an exercise rather than as a substitute for due diligence.

Fifth, the comparison in section 7 rests on documentary evidence of uneven depth for the five countries. It is designed to identify the design features that distinguish their frameworks and to place Pakistan among them, not to rank them precisely, and its scores should be read with the evidence in Appendix C.

3 The Policy Architecture

Six instruments govern Pakistan's electric vehicle and battery sector, and they occupy materially different positions on the path from working group to gazetted law. This section examines each in turn and establishes on which of them an investor can currently rely. The four incentive lines proposed for battery manufacturing, which circulate alongside these instruments, are classified separately in section 3.3 and Appendix A, because they are proposed components of an unfinished instrument rather than instruments in their own right.

3.1 The Instruments That Actually Exist, and Which Ones Bind

Discussion of Pakistani electric vehicle policy tends to focus almost entirely on the National Lithium-Ion Battery Manufacturing Policy, which remains unfinished, while overlooking the fact that Pakistan already has a notified electric vehicle policy that has been in force for a year. The New Energy Vehicle Policy 2025-30 was launched in June 2025⁴ and approved by the Economic Coordination Committee in August 2025.⁵ It is not a draft, not a consultation document and not a pitchbook. It is government policy with a legal funding mechanism attached, and anyone assessing Pakistan's seriousness about vehicle electrification should begin here rather than with the battery policy that has attracted more commentary.

The policy sets a headline target of 30% of new vehicle sales being electric by 2030, rising to 50% by 2040 in the approved version.⁶ Earlier drafts of the framework, as reported at the time, proposed a more aggressive 90% adoption by 2040.⁷ That downward revision between draft and approval is itself a small but

instructive illustration of a point this report returns to repeatedly: targets circulating during consultation are routinely softened before they become policy.

The instrument was developed through consultation with more than 60 experts, institutions and industry stakeholders under a steering committee within the Ministry of Industries and Production established in September 2024, and it carries a governance structure that is unusually specific by the standards of Pakistani industrial policy, with monthly and quarterly review meetings conducted by the steering committee and a performance audit by the Auditor General of Pakistan every six months.⁴

What makes this policy different from the aspirational documents that surround it is that it has a legally sourced revenue stream. Its incentives, including the Pakistan Accelerated Vehicle Electrification programme, universally abbreviated PAVE, are financed through the New Energy Vehicle Adoption Levy, imposed under the Finance Act 2025 at 1% to 3% of the value of new petrol and diesel vehicles, rising with engine capacity, on locally manufactured and imported vehicles alike.^{6,8} This is a design of a kind that development economists generally approve of, taxing the purchase that policy is trying to discourage and recycling the proceeds into the substitute. It is intended to spare the subsidy programme an annual contest against health, education or defence for a discretionary budget allocation, which in the Pakistani fiscal context would be a considerable structural advantage; whether the levy proceeds are ring-fenced for the programme in practice is not established from the public record.

The subsidy schedule underneath the policy is specific enough to be tested. Electric two-wheelers receive up to Rs. 65,000, electric three-wheelers up to Rs. 400,000, and four-wheelers Rs. 15,000 per kWh of battery capacity, alongside free registration for eligible vehicles.⁹ The initial allocation was Rs. 9 billion for financial year 2026, covering a full year target of roughly 116,000 electric motorcycles and 3,170 rickshaws and loaders, or 119,170 vehicles, of which 41,000 were allocated to a pilot phase.¹⁰ This report refers to that phase as the pilot and to the phase that followed it as the second phase. The deliberate weighting toward two- and three-wheelers rather than passenger cars is the single most sensible feature of the entire framework, and it deserves more attention than it has received. Over 90% of parts for two- and three-wheelers are already manufactured locally in Pakistan,⁴ which means subsidy money spent in this segment flows into an existing domestic supply chain rather than leaking straight back out as imports, and it targets the vehicle categories that account for the largest share of daily urban trips and fuel consumption.

Set against this notified and funded instrument, the other five instruments governing the sector occupy materially weaker or different positions. The tariff structure for completely knocked down (CKD) vehicle assembly is notified and in force, but was inherited from an expired policy and now operates on extension. The National Lithium-Ion Battery Manufacturing Policy stands in principle: its shape and chemistry have been agreed, and it awaits the National Tariff Board, the Prime Minister and cabinet. The EV Battery Policy remains in draft. The successor auto policy received the Prime Minister’s approval in principle on 9 September 2026 but has not passed the Economic Coordination Committee or the federal cabinet. Figure 1 plots the six instruments against the distance each has actually travelled, and the resulting distribution warrants close attention before the remainder of this report is considered. Every instrument that would govern battery manufacturing specifically remains unfinished.

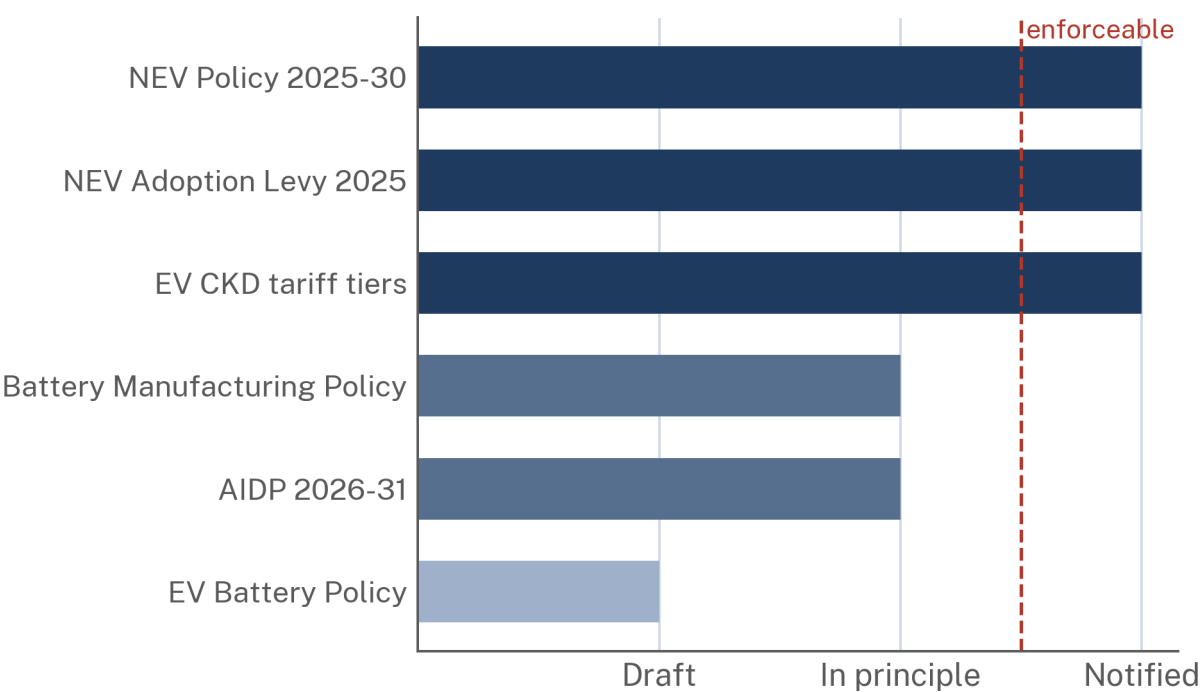


Figure 1. Only three of the six instruments governing the sector are enforceable, and none of the three governs battery manufacturing

Legal standing of the six governing instruments, Pakistan, at September 2026
Source: ARS analysis of Ministry of Industries and Production,⁶ Press Information Department,⁴ Profit by Pakistan Today¹¹ and Geo News.¹²
Notes: Standing as defined in section 2.2 and set out instrument by instrument in Appendix A.

3.2 What PAVE Reveals About Execution Risk

The principal value of a notified policy with a year of implementation behind it lies in what no amount of analysis of draft documents can supply, namely direct evidence as to whether Pakistani institutions are capable of delivering what they announce. Measured against that standard, the record is not encouraging. It also

constitutes the most significant single body of evidence in this report for an investor seeking to price execution risk, for the straightforward reason that it is the only track record available.

The PAVE pilot targeted 41,000 vehicles. By the time the Economic Coordination Committee reviewed the programme in May 2026, 5,409 electric bikes and rickshaws had actually reached recipients.¹⁰ That represents roughly 13% of the pilot target, despite a budget allocation of Rs. 9 billion, political priority, a public launch with considerable fanfare and administration by an established body in the Engineering Development Board. Measured against the full year target of 119,170 vehicles set out in section 3.1, the pilot delivered about 4.5%, the figure officials themselves used when redesigning the scheme. The two ratios describe the same shortfall against different bases. This report uses 13% because it measures the scheme against the target it was actually given, and treats 4.5% as the measure against the year's ambition. Figure 2 shows the shortfall against the revised second-phase target for context.

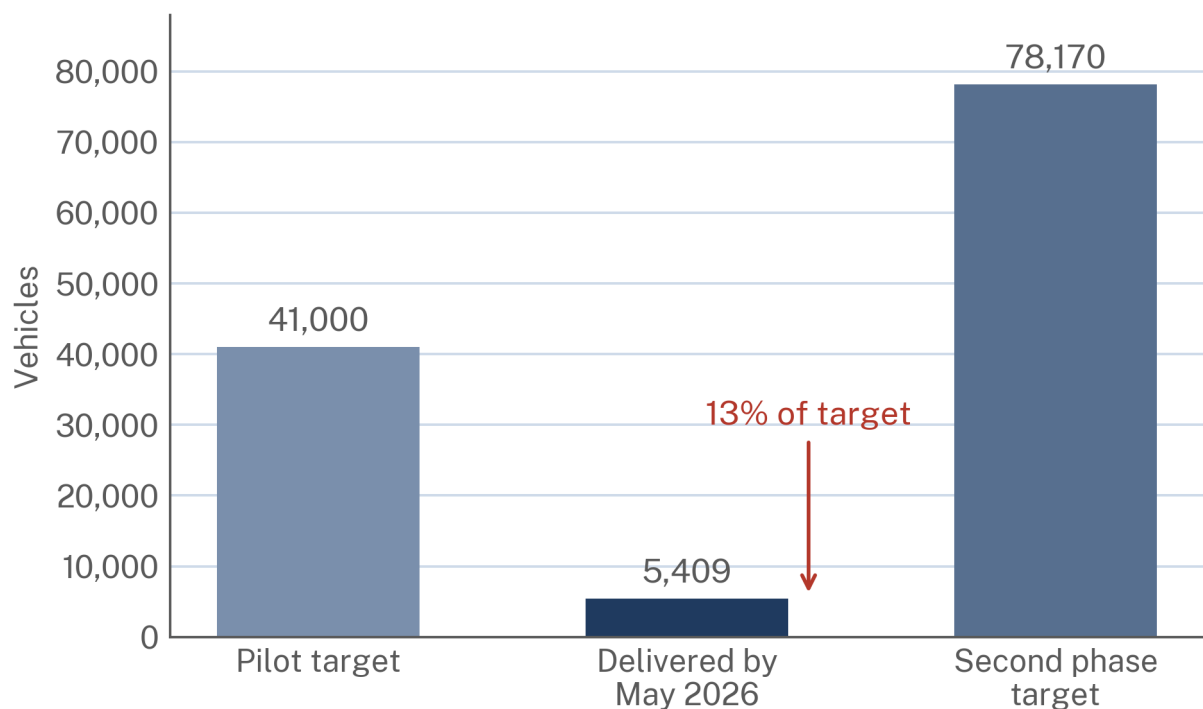


Figure 2. A subsidy scheme with a Rs. 9 billion allocation and political priority delivered 13% of its pilot target

PAVE vehicles targeted and delivered, Pakistan, pilot to May 2026 and second phase

Source: ARS analysis of Profit by Pakistan Today¹⁰ and ProPakistani.¹³

Notes: The second phase target of 78,170 is the balance of the full year target of 119,170 after the 41,000 vehicle pilot.

ARS ASSESSMENT

The gap between announcement and delivery is the most reliable single indicator of execution risk in this sector. On the evidence of the pilot, headline targets in Pakistani industrial policy should be discounted by a substantial and empirically grounded factor rather than taken at face value.

What makes this shortfall genuinely instructive rather than simply disappointing is the diagnosis. The bottleneck was neither citizen demand, which by all accounts substantially exceeded the available allocation, nor manufacturer capacity, since local two-wheeler assemblers had ample ability to supply. It was bank approval. Of 44,689 applications forwarded to banks for lease financing, 9,889 were processed and 4,075 approved.¹⁴ The original scheme design required buyers to pay the full purchase price upfront and then claim reimbursement of the subsidy afterwards, a structure that pushed the entire financing burden onto applicants who were, almost by definition, the low income buyers least able to bridge it and least likely to secure quick bank facilities. The programme failed at the point where policy design met financial constraints.

The government's response to that failure is, to its credit, an unusually direct fix rather than a rhetorical response. The second phase, approved by the Economic Coordination Committee in May 2026 with a Rs. 9 billion package, reverses the payment flow entirely.¹³ Buyers now pay the net price after subsidy deduction directly to the manufacturer, and the Engineering Development Board transfers the subsidy amount to the manufacturer through the State Bank of Pakistan once delivery and registration are verified. The allocation mechanism has also shifted from a computerised lucky draw to first come, first served, removing a layer of delay and opacity. A revised self-financing scheme for federal employees in grade 16 and below requires a down payment of only Rs. 10,000 for a bike. The second phase target has been set at 78,170 vehicles, which is exactly the balance of the full year target after the pilot.

For a Chinese investor, three separate lessons follow from this episode, and they generalise well beyond the two-wheeler segment. The first is that Pakistani policy design has a demonstrated tendency to place financing friction on the party least equipped to absorb it, which is a risk worth checking explicitly for a project with an incentive scheme. The second, more encouraging, is that the system corrected itself within roughly nine months once the failure became visible, which is faster than many comparable jurisdictions manage and suggests the review architecture built into the policy is functioning rather than decorative. The third is that headline targets in Pakistani industrial policy should be discounted

by a substantial and empirically grounded factor rather than taken at face value. If a subsidy scheme with its budget allocated and political priority behind it delivered 13% of its pilot target, then the 3,000 charging station target discussed in section 4.4, and the localisation percentages discussed in section 3.6, deserve to be read through the same lens.

3.3 The National Lithium-Ion Battery Manufacturing Policy

The battery policy has a precisely datable origin, which is unusual enough in Pakistani policymaking to be worth noting on its own. A dedicated working group was constituted on 12 December 2025. Within weeks, a high-level meeting chaired by Haroon Akhtar Khan, the Special Assistant to the Prime Minister on Industries and Production, had approved lithium iron phosphate (LFP) as the chemistry to be prioritised for first-phase localisation.¹¹ By April 2026 the policy had been forwarded to the Ministry of Industries and Production and was awaiting consideration by the National Tariff Board, after which it would go to the Prime Minister and then to cabinet.¹⁵ As of the date of this report it had not completed that journey.

The chemistry decision merits greater recognition than it has generally received, because it is the clearest part of the framework to reflect strategic reasoning rather than aspiration. LFP cells are cheaper than the nickel manganese cobalt cells that dominate premium vehicles and are considerably more thermally stable, which matters in a country with Pakistan's ambient temperatures and its uneven record on industrial safety enforcement. The chemistry is also better suited to the stationary storage applications that Pakistan's solar boom has generated. Most importantly, the two elements that give the chemistry its name are the two raw materials for which Pakistan can most plausibly claim a domestic base. The selection of a chemistry matched to domestic geology, rather than to the global premium standard, is precisely the decision a country in Pakistan's position ought to be taking, and it compares favourably with jurisdictions that have pursued the highest specification chemistry without any corresponding raw material logic.

Around that chemistry decision, the policy is built on three stated pillars: phased localisation, tariff rationalisation and performance-linked incentives. The localisation roadmap that has circulated alongside it targets at least 70% domestic value addition in a first phase from 2026 to 2028, rising to at least 80% from 2029 to 2031.¹⁶ These figures appear consistently across reporting, and they are the numbers most frequently quoted back as though they were policy. They are not. They are targets described in an investment pitchbook and in

ministerial briefings, and in July 2026 the Engineering Development Board's chief executive said the policy still required the Prime Minister's approval before it could be announced.¹⁷ Until the instrument clears the National Tariff Board and the cabinet, it remains at the in principle stage defined in section 2.2: an agreed shape, not an obligation to which a manufacturer can be held or a concession on which a manufacturer can rely.

There is a further structural complication that most coverage misses entirely. The battery policy is not a single document awaiting one signature. It is an umbrella containing at least three work streams running on separate clocks. The localisation and tariff framework itself moved fastest. The Pakistan Standards and Quality Control Authority (PSQCA) testing, certification and recycling regime, which the policy assigns as a precondition, was still being developed as of mid-August 2026, when the Ministry of Industries instructed PSQCA to draw up lithium-ion battery standards; the broader Next Generation Energy Storage Policy that is meant to sit above both was, in the same reporting, awaiting presentation to stakeholders.¹⁸ An investor evaluating a battery project in Pakistan today is evaluating something whose eventual tariff treatment depends on how and when three separate tracks converge, and no public document specifies which takes precedence where they conflict.

Figure 3 summarises the practical consequence. Of the six instruments, three are notified and enforceable, two rest on agreement in principle and one exists only in draft, and none of the enforceable three governs battery manufacturing. All four incentive lines proposed for battery manufacturing exist only in the Engineering Development Board's investment pitchbook.¹⁶ That distribution, rather than any individual headline target, is the honest summary of the ground on which an investor can currently build a financial model.

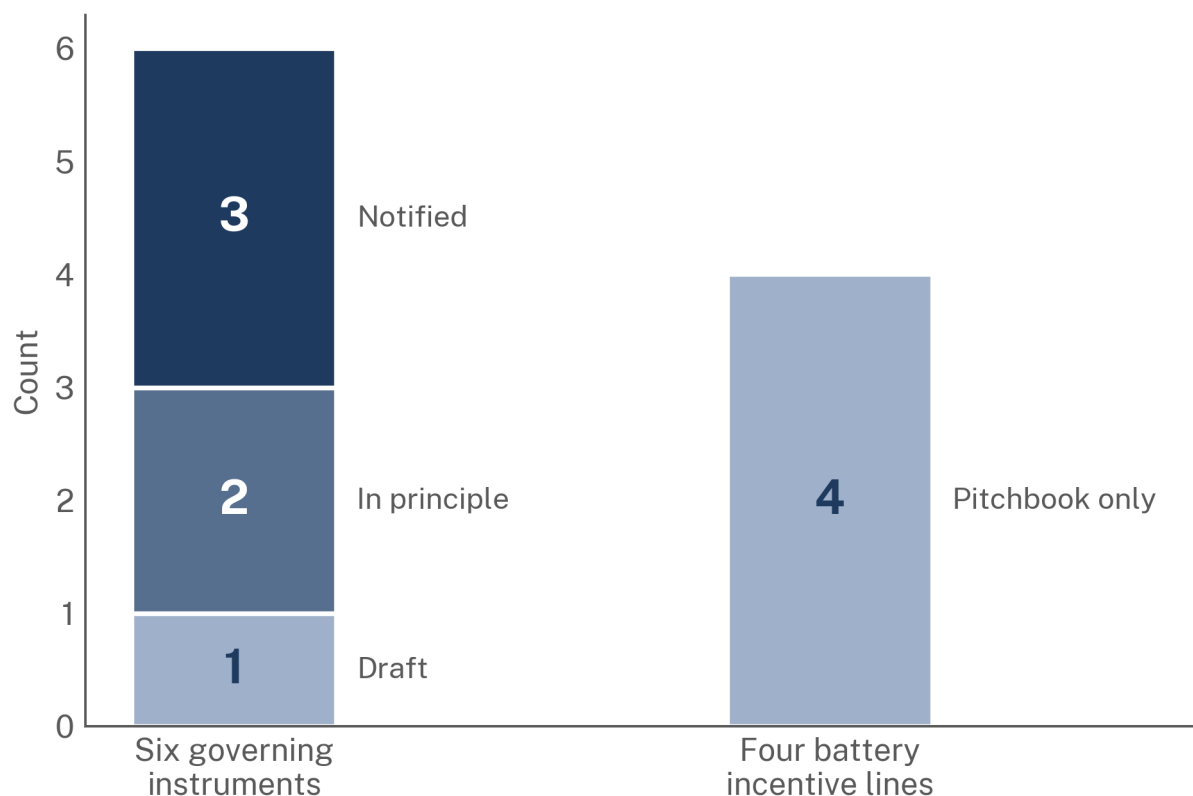


Figure 3. Three of the six governing instruments are enforceable, and all four battery incentive lines exist only in a pitchbook

Governing instruments and battery incentive lines by legal standing, September 2026

Source: ARS analysis of the instruments listed in Appendix A.

Notes: Instruments and incentive lines are counted separately because the incentive lines are proposed components of an unfinished instrument.

3.4 The Tariff Schedule

Separating the enforceable from the announced matters most acutely on tariffs, because tariffs are where an industrial project's economics are actually decided. On the vehicle side, Pakistan already has a functioning notified structure, inherited from the expired auto policy and currently running on extension to June 2027. Components specific to electric vehicles imported for completely knocked down assembly carry a duty of 1%.¹⁹ Non-localised completely knocked down parts, meaning components a manufacturer imports where a domestic supplier exists, carry 10% duty. Parts that have been localised within an approved framework carry 25% duty if imported rather than sourced domestically. The 2026-27 budget extended the incentive on completely knocked down kits for electric vehicles to 30 June 2027 on the same tiers.²⁰

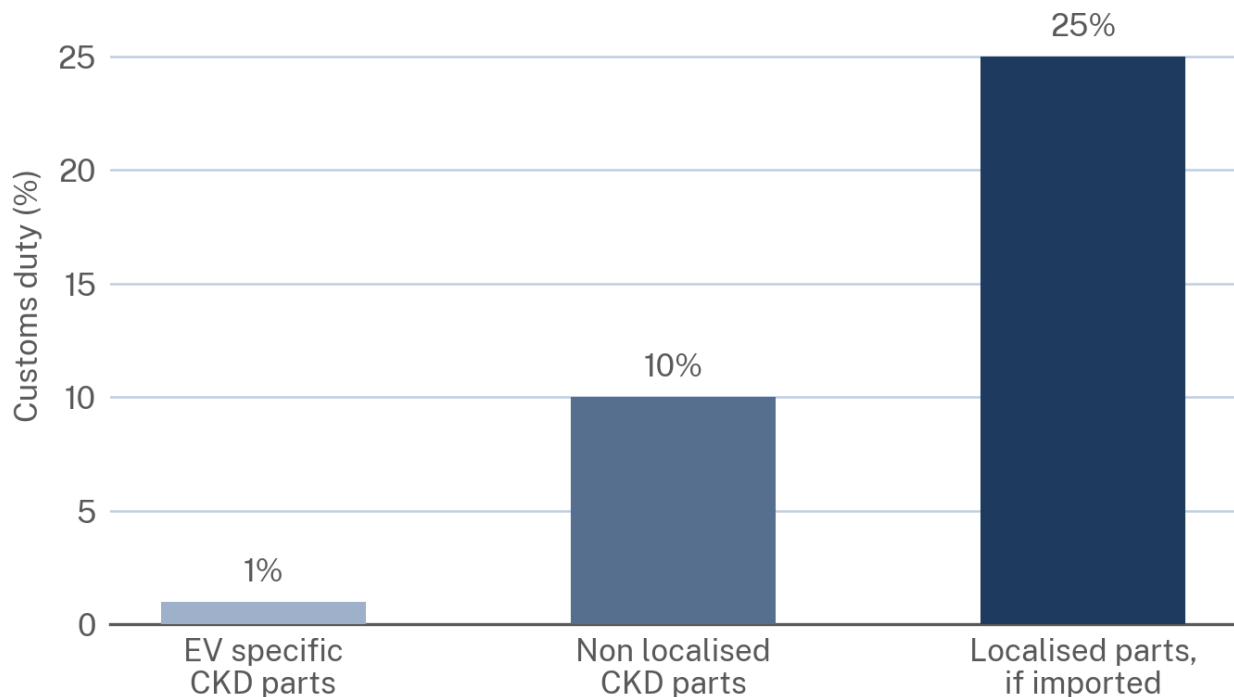


Figure 4. The notified tariff inverts the usual structure: the more a part could be made locally, the dearer it is to import

Customs duty on CKD inputs by localisation status, Pakistan, in force to June 2027

Source: ARS analysis of the completely knocked down tariff tiers inherited from the Auto Industry Development and Export Policy 2021-26 and extended to June 2027.

That inversion is the most sophisticated feature of the existing framework and warrants precise understanding. Under a conventional tariff structure a manufacturer pays progressively less duty as the imported item becomes more processed, which rewards the importation of finished goods. Under the Pakistani structure, the penalty rises once a part is added to the list of localised components, so that a manufacturer continuing to import an item domestic suppliers have begun producing incurs an escalating cost. The schedule of localised parts is updated through periodic revision of the relevant statutory regulatory order (SRO), with the effect that the penalty perimeter expands as domestic capability develops. This represents a genuine industrial policy framework rather than protectionism framed as strategy, with the force of formally notified law rather than merely stated policy intent.

On the battery side specifically, none of the equivalent figures has that status. A 0% duty on imported cells through 2028, a 20% federal excise duty on imported completely built battery units, a 1% sales tax for local battery manufacturers and accelerated depreciation of up to 90% in the first year all appear in the Engineering Development Board's investment pitchbook for battery manufacturing¹⁶ and in ministerial briefings. They describe where the government wants to end up. None can currently be relied upon as an enforceable right, and a project

whose financial model assumes a 0% cell import duty that has not been gazetted is carrying a policy risk quite separate from the ordinary commercial risk of operating in Pakistan.

One narrow concession does have notified status and is worth flagging precisely because of what its survival reveals about government preference. When the previous auto policy expired at the end of June 2026 without a successor in place, the government extended, rather than allowed to lapse, a 1% sales tax concession specifically for locally assembled electric vehicles with battery capacity up to 50 kWh, and light commercial electric vehicles up to 150 kWh, through June 2027.²¹ Incentives for hybrid vehicles were not similarly preserved. Sales tax on hybrid vehicles rose to 25% from 1 July 2026, replacing concessional rates of 8.5% to 12%.²² On 13 September 2026 the rate for locally manufactured hybrids up to 2,000cc was cut back to 18%.²¹ Pure electric vehicles below a defined battery threshold kept their 1% rate throughout, while hybrids lost most of their concession and then recovered part of that concession. That divergence identifies the government's revealed preference with considerably more reliability than any pitchbook statement, and its partial reversal within ten weeks shows how quickly a concession can move.

There is a further tariff development that cuts against the localisation narrative and deserves mention because it has generated real industry anger. A broader tariff rationalisation took effect at almost exactly the moment the old auto policy expired, and industry figures have argued publicly that the resulting structure is import-biased in a way that rewards trading activity over domestic value addition.²³ One senior industry finance executive put the criticism directly, noting that at a moment when Pakistan urgently needs industrialisation, foreign investment, exports and productive employment, a tariff structure tilted toward imports risks rewarding precisely the opposite behaviour. Whether that criticism is fully fair is contestable, since tariff rationalisation has independent macroeconomic justifications tied to Pakistan's commitments under its external financing arrangements, but the timing has left manufacturers without a coherent roadmap at exactly the point when several were making long-term capital commitments.

3.5 The EV Battery Policy

A separate instrument, generally called the EV Battery Policy, is being developed alongside the battery manufacturing policy, and the two are conflated in press coverage often enough that untangling them is worthwhile. What can be stated with confidence is that the government has repeatedly and publicly

committed to finalising a dedicated policy covering electric vehicle batteries and adjacent energy storage, that officials at Special Assistant level have described this as an active priority tied explicitly to the national energy security framework, and that LFP has been confirmed as the preferred choice across both vehicle and stationary storage applications. Officials have also stated an intention to formally follow up on memoranda already signed by Chinese firms, which signals that the government views these agreements as a pipeline to be converted rather than a public relations exercise.

What remains unresolved is everything that would allow a project to be priced. The detailed incentive schedule has not been published in final form. The rollout timeline exists only as an undertaking to finalise the policy soon, a formulation that in the recent history of Pakistani industrial policy has meant anywhere from months to several years. Most consequentially for anyone structuring a transaction today, no public document specifies how the EV Battery Policy will interlock with the National Lithium-Ion Battery Manufacturing Policy, which of them takes precedence where they overlap, or how conflicts between them will be resolved. As of mid-August 2026 the Ministry was still developing the underlying battery standards while the energy storage policy above it was still being finalised, confirming that this is a live process rather than a text awaiting signature.

The right response to this is not to avoid the sector but to structure around the uncertainty rather than assume it away. A joint venture agreement that fixes a specific tariff or tax rate as a contractual term, on the assumption that the eventual policy will simply mirror the current pitchbook, is exposed to a risk neither party controls. The more defensible structure stages the investment against policy milestones. An initial tranche can proceed on the basis of what is already notified, covering pack assembly, battery management system integration and testing infrastructure, none of which depend on the unfinished cell-level schedule. Later tranches, particularly anything committing to cell manufacturing or upstream material processing, should be conditioned on formal gazetting and should carry a defined renegotiation mechanism if the notified terms diverge materially from what was described during negotiation.

It is worth being explicit about what would count as genuine confirmation, so that future developments can be measured against a fixed bar rather than another round of statements. Three things: a gazette notification of the finalised policy, publication of the PSQCA battery safety and certification standards in final form, and a published tariff schedule specific to cells, cathode materials, anode materials and finished packs that supersedes the pitchbook figures. Until all

three have occurred, this instrument should remain classified as draft, in the terms of section 2.2. That is not scepticism about Pakistan's seriousness. It is the same discipline any competent adviser would apply to a draft regulation in any jurisdiction, and Pakistan's automotive sector has a documented history of policy documents circulating in draft for years before notification, or lapsing entirely and being replaced by a differently structured successor.

There is one asymmetry in this uncertainty that favours an early mover, and it deserves to be acknowledged rather than being buried under caution. Because the incentive schedule is unfinished, it is also still open to influence. The broader policy process has explicitly involved industry consultation, with more than 60 experts, institutions and industry stakeholders consulted on the parallel electric vehicle policy.⁴ A Chinese manufacturer with genuine capital at stake and a credible commitment to local value addition is in a materially better position to shape the final text than one arriving after gazettal, when the terms are fixed and the only remaining question is compliance. Investors who read regulatory uncertainty exclusively as risk sometimes miss that in a jurisdiction actively drafting, uncertainty is also access and an opportunity.

3.6 Local Content and the Successor Auto Policy

The Auto Industry Development and Export Policy 2021-26 expired on 30 June 2026, leaving the sector without a governing framework while its successor remained under negotiation. Under the expired policy the local value addition requirement for parts manufactured under SRO 655(I)/2006 was fixed at 30%, excluding engines, gearboxes and transmission parts.¹⁹ In the two and three wheeler segment local content now exceeds 90%.⁴

That 30% floor is the right anchor against which to read what is now proposed, and the proposal has already moved once. In August 2026 a steering committee reportedly led by Ishaq Dar reached consensus in principle on a successor under which cars below 1000cc would reach 80% domestic value addition by financial year 2031 and sport utility vehicles 55%,²⁴ with the tariff premium on imported vehicles, reported at around 40%, falling to zero by financial year 2030.²⁵ On 9 September 2026 the Prime Minister gave approval in principle to the Automotive Industry Development Policy 2026-31 on different terms: mandatory domestic value addition reaching 40% for passenger cars, 45% for commercial vehicles and 90% for motorcycles and rickshaws by financial year 2031, a weighted average import tariff falling from 15.7% to 5.99% by 2030, a 12% export performance requirement for passenger car makers and a Rs. 100.36 billion electrification programme.¹² The policy still requires review by the Economic Coordination

Committee, the cabinet and the International Monetary Fund before it can be notified.²⁶ This report adopts the September terms as the later and more senior approval, and reports the August figures because they remain the ones most widely quoted.

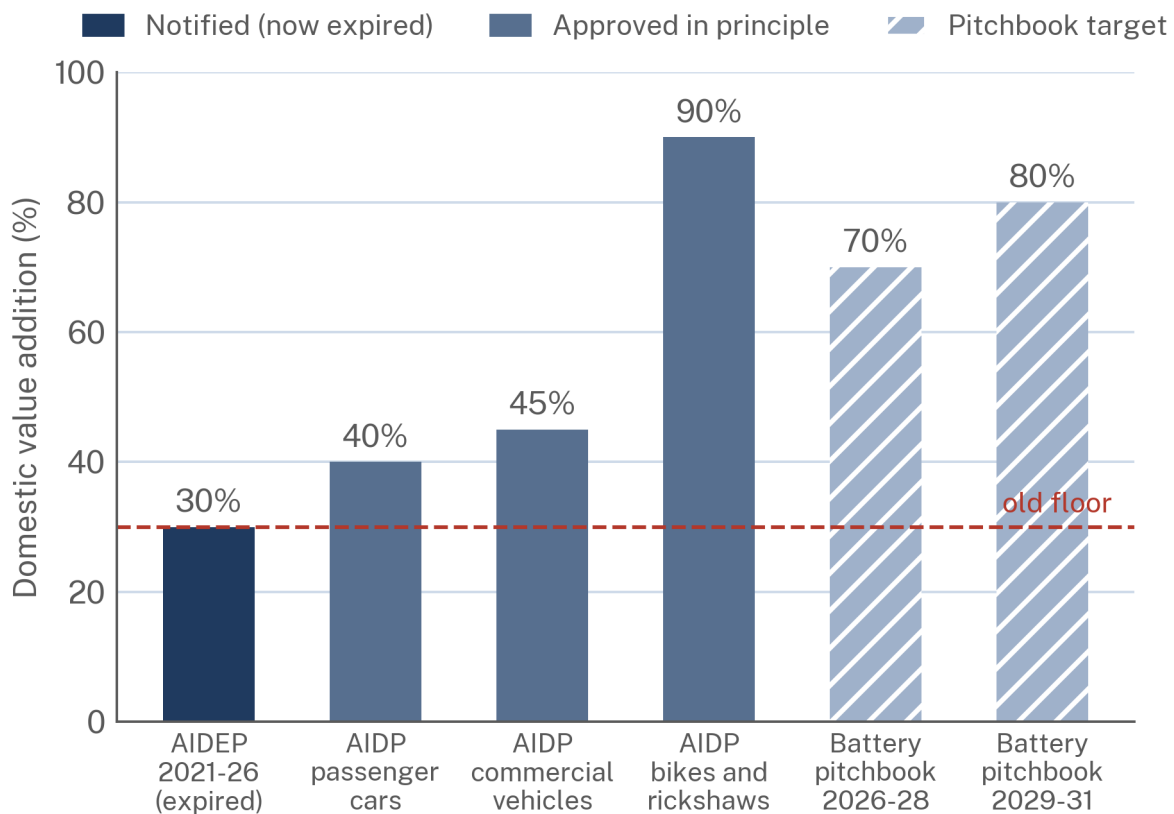


Figure 5. The approved successor raises the car threshold only modestly, while the battery pitchbook targets sit far above it

Domestic value addition requirements and targets by instrument, Pakistan, expired policy against current proposals

Source: ARS analysis of Engineering Development Board,¹⁹ Geo News¹² and Engineering Development Board.¹⁶

Notes: Battery policy figures are pitchbook targets and carry no legal force. The August 2026 steering committee proposal of 80% for small cars and 55% for sport utility vehicles was superseded by the September terms shown.

The September terms change the character of the successor policy. The passenger car threshold rises only from 30% to 40%, a far smaller step than the August proposal, while tariff protection still falls substantially. The design continues to penalise operations that remain limited to assembly and to reward manufacturers that build domestic supply chains, but the softening between August and September is itself the pattern documented in section 3.1: targets circulating during consultation are revised before they become policy. The 90% threshold for motorcycles and rickshaws, by contrast, largely codifies what the two and three wheeler industry already achieves.

Neither version is notified law. The September approval is more senior than the August consensus, but it remains an approval in principle, and the percentages should be read as the current proposal rather than settled fact. A prudent model should use a range rather than a point estimate, and the revision between August and September already shows the direction in which such ranges tend to move.

For due diligence, it is more useful to disaggregate local content layer by layer than to track a headline percentage, because a single blended number conceals exactly the substitution this report warns about throughout. Figure 6 sets out the seven layers together with this report's assessment of where Pakistani capability presently sits in each. The figure establishes that an 80% headline could in principle be attained almost entirely through body, chassis, wiring and thermal systems, the two layers in which Pakistan already possesses decades of experience, while the motor, the battery management system and the cells themselves remain wholly import dependent. Under the September terms the passenger car threshold is 40%, which body and chassis work alone could approach, so for cars the risk is a threshold met with no electric drivetrain content at all; the layering matters most for the 90% threshold for electric motorcycles and rickshaws and for the 70% to 80% battery targets. Any assessment of a particular venture's compliance should therefore require this breakdown rather than accept an aggregate figure, since the aggregate discloses almost nothing as to whether the strategically significant components of the vehicle are being produced in Pakistan.

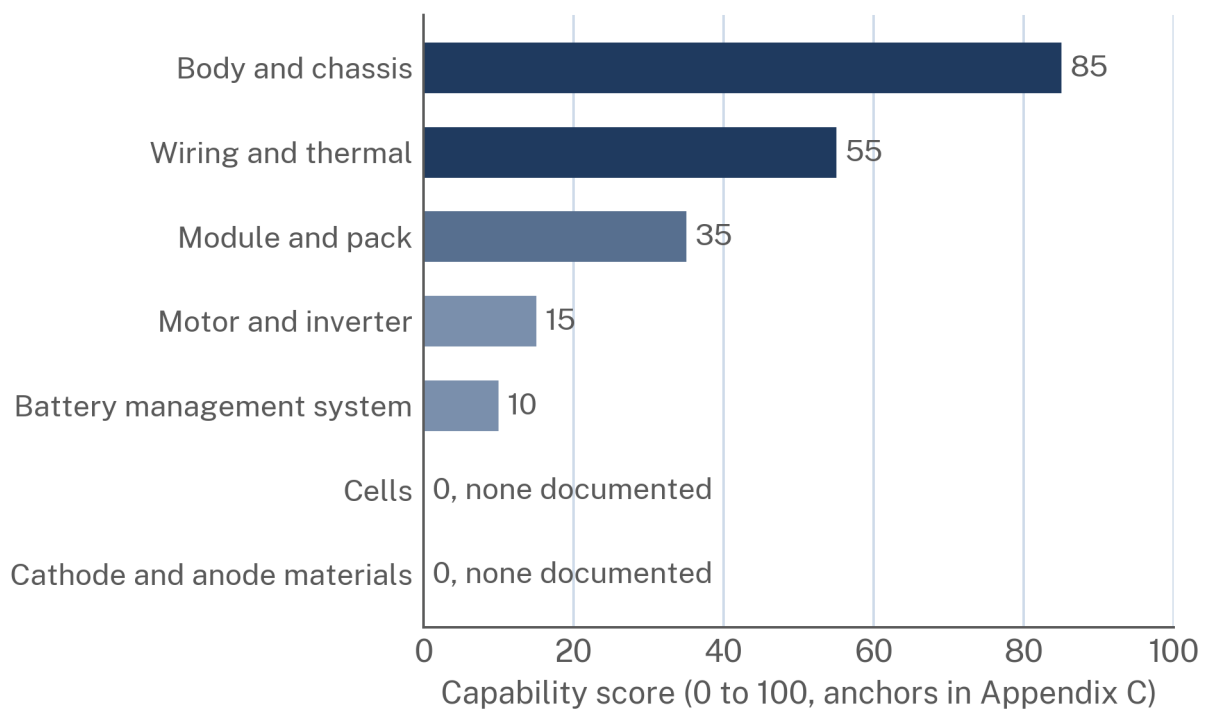


Figure 6. An 80% headline figure can be reached without touching a single strategically significant component

Indicative Pakistani capability by vehicle and battery value chain layer, ARS analytical scoring, September 2026

Source: ARS analysis of the evidence reviewed in sections 3 to 6.

Notes: Analytical scoring, not a measured production share. Scores are ordinal and rank the layers against one another on the evidence reviewed.

4 The Market

The market evidence divides into four questions: what the sales data actually shows, where Chinese vehicles sit on the Pakistani price ladder, whether the running cost advantage survives examination, and whether the charging network can support the demand the rest of the framework is trying to create.

4.1 The 61% Jump, Examined

The most widely circulated statistic about Pakistan's electric vehicle market this year is that sales rose 61% in a single month. The figure is accurate. It originates in Arif Habib Limited's monthly automotive tracking and was subsequently reported by ProPakistani and several other outlets.²⁷ It describes passenger electric vehicle sales moving from 33 units in February 2026 to 53 units in March 2026.

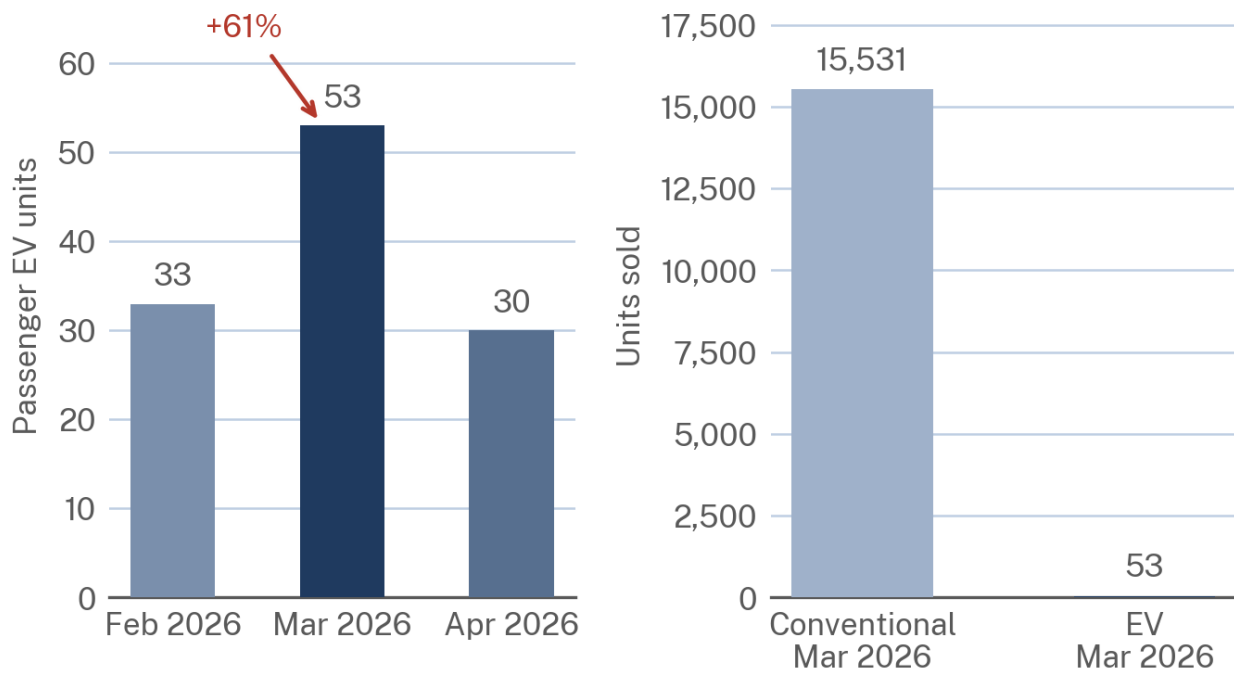


Figure 7. The 61% jump describes twenty additional vehicles in a market that sold 15,531 conventional units the same month

Passenger electric vehicle units sold, February to April 2026, against conventional vehicle sales for March 2026

Source: ARS analysis of Pakistan Automotive Manufacturers Association data and Arif Habib Limited monthly tracking as reported in ProPakistani²⁷ and PakWheels.²⁸

Notes: Both panels show the same March 2026 data. The right panel adds the denominator that most coverage omits. In the April model level data the Honri Ve is the only electric model listed, so the electric series is in effect one model's sales.

53 units is not a market. It is a rounding error against the roughly 15,500 conventional cars, light commercial vehicles, vans and jeeps sold in the same month. The percentage is real and the direction it indicates is real, but its reproduction without the underlying base conveys to an investor a materially inflated impression of present demand. The figure has been cited in investment contexts as though it described growth in market share rather than the sale of 20 additional vehicles.

The surrounding context makes the picture more complicated. March 2026 was a weak month for conventional vehicles across the board. Petrol car sales fell 9% month on month,²⁷ with the 1000cc segment, historically the backbone of the Pakistani market, dropping by 36%. Cars below 1000cc slipped 18% and even the 1300cc and above category declined by 6%. Against that backdrop, a small absolute increase in a statistically negligible segment looks less like electric vehicles pulling ahead on their merits and more like one soft month for the broader market coinciding with modest movement in a category too small to support conclusions. It is also important to note that, despite the month on month weakness,

overall vehicle sales were still 40% higher year on year, indicating that the broader market was undergoing a genuine recovery.

April made the trend claim harder still. In the model level data for April, the Honri Ve is the only electric model listed, and it fell 43% month on month, from 53 units to 30.²⁸ The reported monthly electric series is therefore, in effect, the sales of a single model, and no BYD model appears in the series. Whether the fall represents a genuine reversal, ordinary volatility in a market where a single fleet order swings the percentage dramatically, or a reporting artefact cannot be determined from public data. The defensible conclusion is that March confirmed the existence of a real demand signal and that the public dataset is too fragmented to support any claim about trajectory beyond its existence.

The fragmentation is structural rather than temporary, and it is worth understanding because it affects every number in this section. Pakistan has no single centralised dataset combining monthly electric, hybrid and plug-in hybrid registrations by brand and province. The Pakistan Automotive Manufacturers Association bulletins, the most cited source, are built around member manufacturers reporting largely conventional vehicles, with electric figures appearing inconsistently and sometimes aggregated in ways that defeat brand level comparison across months. Provincial registration authorities keep their own records in formats that resist aggregation. Customs data captures imported units but is not published broken down by engine capacity or battery powertrain in a form that reconciles with either. Anyone building a national picture has to triangulate across manufacturer disclosures, provincial registrations, customs statistics and company announcements, each capturing a different, partially overlapping slice of the market.

Against the enthusiasm the March figure generated, there is one countervailing data point that deserves attention because it speaks to whether early adoption is converting into durable confidence. Dealer and buyer reporting from the 2025-26 period describes steep early depreciation on some electric models, driven by buyer scepticism about battery degradation and range anxiety in a market with no mature secondary market for electric vehicles and no standardised battery health certification. Pakistani car buying culture treats a vehicle as a liquid financial asset with an established resale market, an expectation electric vehicles do not yet meet. No systematic resale data are published, so the size of the effect cannot be stated with confidence. This is a genuine demand side headwind sitting alongside the more commonly discussed constraints of price and charging access, and it receives less attention than it deserves.

4.2 Where Chinese Vehicles Actually Sit on the Price Ladder

The most useful reference point for understanding Pakistani vehicle pricing is the Suzuki Alto, which has defined the mass market for decades and continues to do so; it was the highest selling individual model in April 2026, at 7,567 units.²⁸ No electric vehicle currently sold in Pakistan competes with the Alto on price, and none is likely to do so at current battery costs. That gap is the clearest available evidence that affordability rather than technology preference remains the binding constraint on mass adoption.

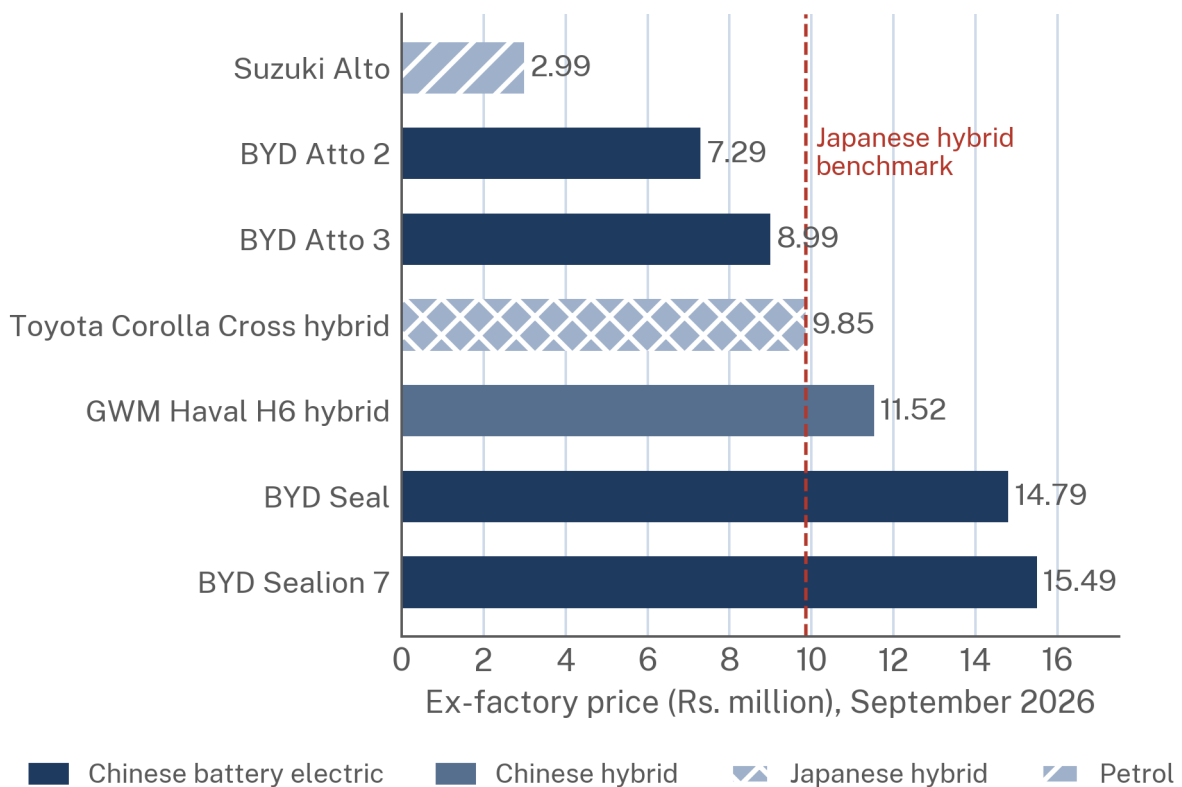


Figure 8. BYD’s two cheapest electric models undercut the Japanese hybrid benchmark, but nothing approaches the segment that drives volume

Ex-factory prices for selected models on sale in Pakistan, Rs. million, September 2026

Source: ARS analysis of manufacturer ex-factory prices as listed by PakWheels.²⁹

Notes: Prices exclude freight, registration, withholding tax and dealer documentation charges, which add materially to the price a buyer pays and widen with vehicle value.

Above the entry tier the ladder has widened genuinely over the past year, from what was effectively a single premium niche into a structured set of segments. BYD’s Atto 2 launched in late January 2026 at roughly Rs. 7,290,000,²⁹ making it the most affordable brand new electric sport utility vehicle on sale and positioning it against petrol powered B segment vehicles such as the Chery Tiggo 4 and MG ZS rather than against other electric vehicles. That targeting choice is strategically significant. Rather than competing for share within the tiny existing

electric buyer pool, the Atto 2 is aimed at conquest sales from petrol buyers who had not previously considered electric at all.

The Atto 3 sits roughly Rs. 1,700,000 above it at approximately Rs. 8,990,000, offering a larger footprint and a cabin that reviewers and owners consistently describe as unusually premium for the bracket, built on the same blade battery architecture BYD uses across its range. Both sit below the locally assembled Toyota Corolla Cross hybrid, which at Rs. 9,849,000 is the natural Japanese benchmark for this segment, and that is new in a market Japanese brands have dominated for decades. The Chinese price advantage does not extend across the range. GWM's Haval H6 hybrid lists at Rs. 11,523,015, above the Toyota, and BYD's Seal and Sealion 7 sit between Rs. 14,790,000 and Rs. 16,990,000.²⁹

What this widening implies for localisation is fairly direct. High volume, more affordable models are the most plausible candidates for local manufacturing at scale, because their volumes can justify the fixed costs of establishing domestic supply for seats, wiring harnesses and body panels. Premium vehicles will likely remain import heavy regardless of what local content thresholds eventually require, because the volumes rarely justify dedicated tooling and manufacturers at that end will accept a tariff penalty rather than invest in low volume local production. This bifurcation appears in virtually every developing automotive market and there is no reason to expect Pakistan to be an exception.

One further dynamic complicates any simple story about Chinese vehicles undercutting on cost. Every price above is an ex factory or listed price point excluding freight, registration, withholding tax and dealer documentation charges, all of which add meaningfully to what a buyer actually pays. Because Pakistani vehicle taxation is structured progressively with respect to value, this wedge widens as prices rise, so the effective landed price diverges from the listed figure most at exactly the end of the market where Chinese manufacturers have been positioning their more profitable models. Any competitive benchmarking that uses ex factory prices alone will systematically overstate how competitively these vehicles are priced by the time they reach a showroom floor.

It is also worth noting what the price ladder does not yet contain, because the absence is strategically significant. There is no locally assembled electric vehicle at a genuinely mass market price point, and there will not be until the Gharo plant discussed in section 6 reaches volume and begins passing its cost advantage through to consumers. The most consequential pricing event in this market over the next 18 months is therefore not a new model launch but the first locally assembled BYD reaching dealers, because that is the point at which the import

duty component of the price comes out and the ladder's bottom rung has a chance of moving down toward where Pakistani buyers actually are.

4.3 The Running Cost Arithmetic

A specific comparison has circulated widely enough in Pakistani marketing and social media to have taken on a life of its own, claiming that charging an electric vehicle costs around Rs. 4,000 a month against a petrol equivalent of roughly Rs. 60,500. This report could not trace the figure to a documented calculation. It is not a methodology, and it should not be treated as representative.

A defensible calculation needs four inputs, and one of them has been widely misread. Petrol sold at Rs. 390.12 per litre from 24 September 2026.³⁰ A reasonably efficient petrol car achieves around 10 kilometres per litre in Pakistani mixed driving, and a comparable electric vehicle consumes approximately 15 kWh per 100 kilometres, consistent with the compact and midsize electric sport utility vehicles actually sold in Pakistan; both are stated assumptions rather than measured figures. The fourth input is the price of electricity at the plug. The figure usually quoted, Rs. 39.70 per kWh, is not that price. It is the tariff, inclusive of tax, that distribution companies charge charging station operators after the National Electric Power Regulatory Authority (NEPRA) cut the base rate to Rs. 23.57 per kWh in April 2025, and NEPRA removed the cap on the operator's margin at the same time, leaving the price to the driver to the market.^{31,32,33} Reported public prices vary widely by operator and charger type. One Lahore guide put public fast charging at about Rs. 45 per kWh in April 2026,³⁴ while a national station directory lists Rs. 40 to 90 per kWh for public AC charging and Rs. 80 to 160 per kWh for DC fast charging in July 2026.³⁵ The two are reported here rather than reconciled.

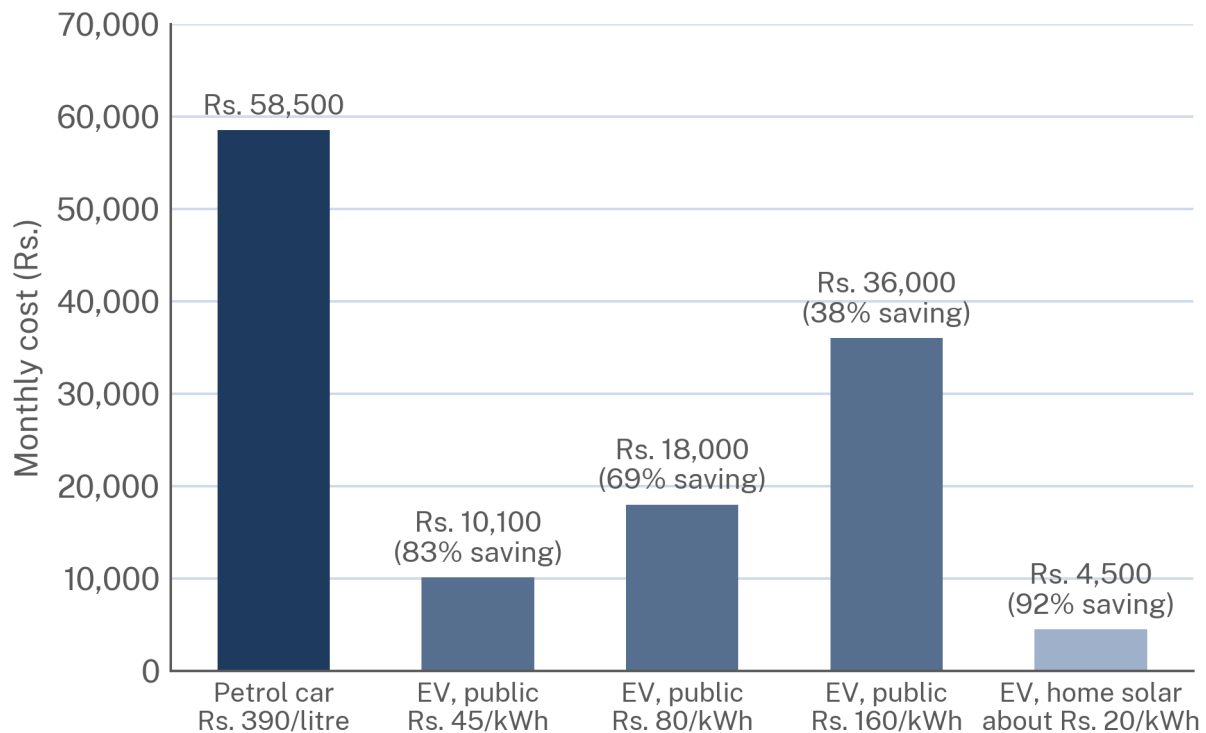


Figure 9. At current prices public charging saves between 38% and 83% against petrol; the widely quoted figure describes home solar charging

Monthly running cost at 1,500 km, Rs., petrol against public and home solar charging, September 2026

Source: ARS analysis of a petrol price of Rs. 390.12 per litre,³⁰ a reported public charging price of Rs. 45 per kWh,³⁴ an intermediate case of Rs. 80 per kWh, the top of the reported fast charging range at Rs. 160 per kWh³⁵ and an indicative home solar rate of Rs. 20 per kWh. Fuel economy of 10 km per litre and electric consumption of 15 kWh per 100 km are assumptions.

Notes: ARS calculation. Every input is named above and the working is retained.

For an active urban commuter in Lahore or Karachi travelling approximately 1,500 kilometres a month, the petrol bill comes to roughly Rs. 58,500. Charged at Rs. 45 per kWh, the same distance costs approximately Rs. 10,100, a saving of about 83%; at Rs. 80 it costs Rs. 18,000, a saving of about 69%; and at Rs. 160, the top of the reported fast charging range, it costs Rs. 36,000, a saving of about 38%. A driver charging at home from rooftop solar at something closer to Rs. 20 per kWh would pay around Rs. 4,500, which is almost certainly where the widely quoted Rs. 4,000 figure originated, as an outcome specific to solar home charging rather than any market average.

That distinction is not a technicality, and it matters more for infrastructure investors than for vehicle buyers. An operator billing at Rs. 45 per kWh earns a margin of about Rs. 5 over its regulated input cost and passes most of the saving to the driver; one billing Rs. 160 keeps most of that saving. Even at the top of the range the driver saves more than a third against petrol, which remains a compelling proposition without any home solar advantage. But how the saving is divided between operator and driver is unsettled, and public charging is a

fundamentally different business, with different revenue per kWh, than a market where most charging happens at home on solar and public infrastructure exists mainly to serve intercity travel. Given Pakistan's solar boom and the strong overlap between rooftop solar adopters and early electric vehicle buyers, it is plausible that a meaningful share of current charging happens at the cheaper home rate, which would mean the aggregate advantage for today's buyer base sits closer to the dramatic end of the range even though the viral framing overstates it as a universal figure.

The regulatory history behind the Rs. 39.70 tariff is itself worth understanding, because it did not emerge from nowhere and its durability is an open question. The tariff rationalisation process for charging stations, documented in regulator and Power Division filings through 2025 and 2026, involved a deliberate discount against the commercial tariff that would otherwise apply to a business drawing that much power, to make public charging viable for operators while keeping the driver facing price low enough to sustain the value proposition. This is a subsidy in substance if not in name, and subsidies in Pakistan's power sector have a well documented history of coming under fiscal pressure.³³ Any long term model for a charging business should stress test what happens to unit economics if that discount narrows or disappears.

There is a national scale version of this arithmetic that policymakers have used to justify the whole programme, and as announced it does not survive the test this report applies elsewhere. The government's announcement of the policy projects total electric vehicle energy demand over the five year policy period at 126 TWh, to be met from existing surplus in the national grid.⁴ Pakistan's grid generated about 137 TWh of electricity in financial year 2024.³⁶ Spread evenly, 126 TWh over five years is about 25 TWh a year, close to a fifth of national grid generation, drawn by a fleet that the same policy expects to reach 30% of new sales only by 2030. The policy document itself resolves the discrepancy. It gives 126.5 TWh as national electricity utilisation in financial year 2024 and projects additional demand from electric vehicles of 1,544 GWh over the policy period;⁶ the announcement appears to have attached the national figure to electric vehicles. The policy's own projection is consistent with a simple cross-check, since two million electric two wheelers, each using an assumed 500 kWh a year, would draw about 1 TWh a year. At that scale charging demand can plausibly be met from existing surplus capacity, so the surplus claim survives, but on the policy document's figure rather than the announcement's.

The same announcement projects a reduction from Rs. 174 billion to Rs. 105 billion, figures often reported as the fuel import bill. The announcement attributes

them instead to capacity payments, the fixed charges paid to power producers whatever their output, on the logic that charging demand would use capacity already paid for.⁴ Against total capacity payments of about Rs. 1.9 trillion in financial year 2024,³⁶ the projected saving is under 4% of the burden. The fuel import saving the announcement claims is separate, at nearly USD 1 billion a year, with carbon credits put at a further Rs. 15 billion. These figures are reported as the government’s projections; the published record does not allow them to be tested.

4.4 Charging Infrastructure as the Binding Constraint

Pakistan has no officially verified count of its public charging network, which is itself informative. In January 2026 the National Energy Efficiency and Conservation Authority described the network as over 20 public charging points in urban centres, and reported 13 licences issued under its 2024 charging regulations.³⁷ Against that base, the national target is 3,000 stations by 2030, with 240 planned for the current financial year, and the electric vehicle policy specifies a corridor design of roughly 40 new stations positioned approximately every 105 kilometres along motorways;⁴ the policy document itself adds a target of one station every 50 kilometres along major highways by 2030.⁶

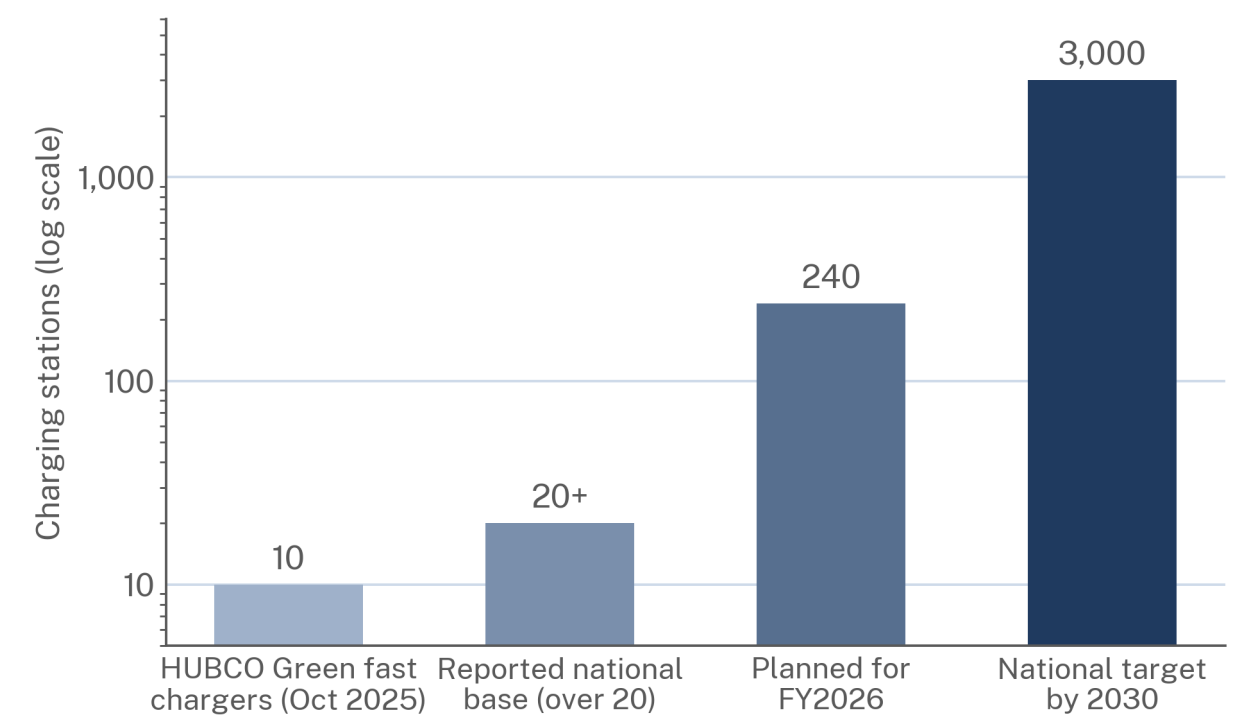


Figure 10. The national charging target sits more than two orders of magnitude above the reported base

Public charging stations, reported base and official targets, Pakistan, logarithmic scale, 2026
Source: ARS analysis of Business Recorder,³⁸ Arab News³⁷ and Press Information Department.⁴

Notes: A logarithmic scale is necessary because the reported base and the 2030 target differ by more than two orders of magnitude. The reported base is the authority's description of over 20 public points in January 2026; the HUBCO Green chargers, counted in October 2025, may be included in that figure.

The geographic distribution of what exists is described inconsistently by different sources, which again reflects the absence of any authoritative dataset. Islamabad and Lahore are generally described as the best served cities. Karachi is described by some sources as having the thinnest public network of the three major cities relative to population and by others as having the largest raw count. Any siting decision premised on Karachi being either underserved or adequately served needs independent verification rather than reliance on published figures.

The development that adds the most substance to an otherwise thin picture is HUBCO Green's motorway corridor, which connects directly to the assembly venture discussed in section 6. HUBCO Green, the group's charging subsidiary, had 10 direct current fast chargers operating in Karachi, Lahore and Islamabad in October 2025, with two more being commissioned and a first corridor phase planned through Hyderabad, Moro, Sukkur and Zahir Pir.³⁸ With PSO and BYD it has announced a network of 128 fast chargers spaced every 150 to 200 kilometres from Karachi to Peshawar.³⁹ This is arguably as strategically important to BYD's Pakistani prospects as the assembly plant itself, because a manufacturer can sell electric vehicles indefinitely without expanding the addressable market if buyers remain confined to short urban trips by charging anxiety. A functioning long distance corridor, even a sparse one, changes the calculation for a buyer considering an electric vehicle as a primary rather than a second car.

The material detail is that the corridor investment originates from the same corporate group constructing the assembly capacity rather than from an unrelated third party. This indicates that at least one major investor in Pakistan has recognised that vehicle sales and charging infrastructure must be developed in parallel rather than sequentially, and it establishes an integrated position that a later entrant would find expensive to replicate. For a Chinese investor assessing entry, the significance is competitive rather than merely infrastructural. The most commercially attractive charging corridor in the country is already controlled by a group holding a specific vehicle brand relationship.

The more useful question than how many chargers exist nationally is how many exist at reasonable intervals along the corridors Pakistanis actually travel, and in the secondary cities outside the three major cities. A fourth charger within an already served Lahore neighbourhood adds little. One installed at a sensible interval on the Karachi to Hyderabad stretch, or in Faisalabad, Multan or Peshawar, does considerably more to make electric vehicle ownership practical

rather than an urban novelty. Given that the 3,000 station target implies an enormous buildout against today's base, the sequencing of that buildout will matter as much as the total, and on the evidence of the delivery record examined in section 3.2, the total itself should be treated as directional rather than as a plan.

5 Supply Chain and Mineral Reality

The policy and market evidence establishes intent and demand. This section establishes capability: where Pakistani industrial activity actually sits on the battery value chain, how deep the import dependency runs, and whether the mineral base can support the chemistry the policy has selected.

5.1 Assembly, Manufacturing, and the Test That Separates Them

The most useful discipline available to an investor assessing any Pakistani battery or vehicle project is a classification test that is quickly applied and that resolves the greater part of the confusion surrounding this sector. If a project imports finished cells and its Pakistani operation builds them into modules and packs, it is assembly and value addition, which is genuine, useful, investable industrial activity, but is categorically not cell manufacturing. If a project makes electrochemical cells from active materials with real local process control, then it is manufacturing in the true sense of the word.

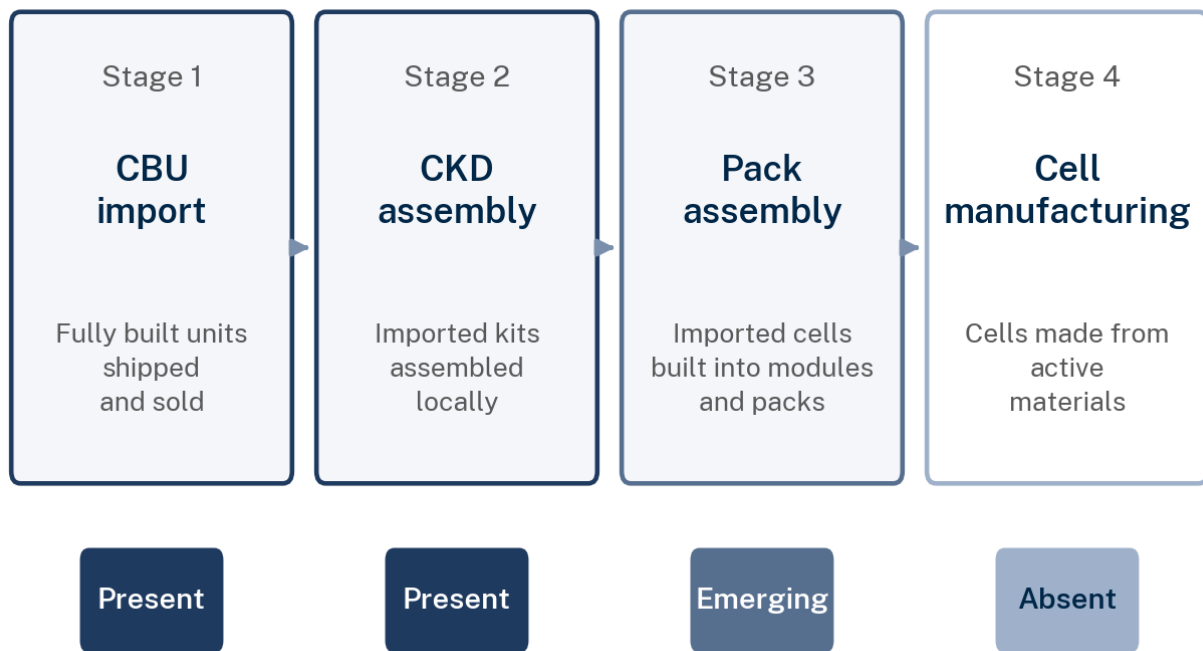


Figure 11. Applied honestly, the test places essentially all current Pakistani activity in the first two stages

The four stages of the battery and vehicle value chain, with Pakistan's present position at each

Source: ARS analysis of the projects and facilities examined in sections 5 and 6.

Notes: Analytical classification. Stage presence denotes documented activity, not installed capacity.

Applied across the current landscape, the evidence overwhelmingly favours the earlier stages. Pakistan has meaningful and growing evidence of pack and system assembly, battery management system integration, imported cells being packaged into finished products, and a rapidly expanding storage distribution network. It has essentially no evidence of cell production. The capabilities that would need to be developed to get there, including cell manufacturing, battery management system design, testing and qualification infrastructure, a functioning recycling loop and upstream material processing, are identified as absent or limited in the government's own investment promotion material,¹⁶ which is a candid admission worth taking seriously rather than dismissing as boilerplate.

The clearest illustration is the country's first dedicated lithium battery assembly facility, in the Korangi Industrial Area of Karachi, developed by the Pakistan Solar Association in coordination with the Engineering Development Board.⁴⁰ Its significance is real. It is the first facility of its kind and represents actual capital deployed into battery assembly rather than vehicle assembly, and it targets the electric two and three wheeler segment, which is where Pakistan has the clearest near term path to meaningful electrification given the price sensitivity documented in section 4.

Its limitations are equally real and rarely stated alongside the enthusiasm. Reporting describes an initial capacity around 4 MW, and separately an output of roughly 2,000 units a month for small vehicle batteries.⁴¹ Both describe a modest operation by any international standard for battery manufacturing, let alone anything resembling the gigafactory concept occasionally used in Pakistan. The facility matters as a first mover and a proof of concept. It does not establish a scaled domestic cell industry, and treating it as evidence that Pakistan has crossed into cell manufacturing would substantially overstate what has been built.

The same terminological slippage runs through the vehicle side. Industry analysts covering the sector have been explicit that under current arrangements Chinese companies are largely importing vehicles and assembling them locally through completely built or semi assembled routes without major investment in localisation or technology transfer. This is not a criticism specific to any manufacturer, and it reflects a rational calculation in a market where demand certainty is limited and the tariff framework is unsettled. But it does mean a facility publicly described as a manufacturing plant may be performing an assembly function adding considerably less domestic value than the word implies, and due diligence should insist on the actual bill of materials and the share sourced domestically rather than accepting a project's self description.

The practical structuring implication is to stage a project's ambitions against the ladder rather than pricing an entire investment as though it will occupy the top rung immediately. A first stage can legitimately and profitably combine imported cells with local module and pack assembly, battery management system integration and testing, all genuine value addition over pure imports and consistent with actual Pakistani capability. A second stage should target local electrode and material processing. A third, conditioned on cell line economics, qualification pathways once standards are finalised, and a genuine recycling loop, would be the point at which an operation could honestly describe itself as cell manufacturing. A project priced at stage 3 valuations while operating at stage 1 should not receive stage 3 policy treatment or investor confidence.

5.2 Import Dependency and What the Trade Data Shows

The scale of Pakistani dependence on imported cells and packs is best understood through the lithium-ion storage import data compiled by Renewables First, an Islamabad think tank whose report *From Solar Panels to Storage: Pakistan's Battery Boom Begins* provides the most detailed public picture available.¹ Cumulative lithium-ion battery imports since 2018 reached 7.6 GWh by the

end of 2025. The trajectory behind that total is what makes it striking rather than merely large.

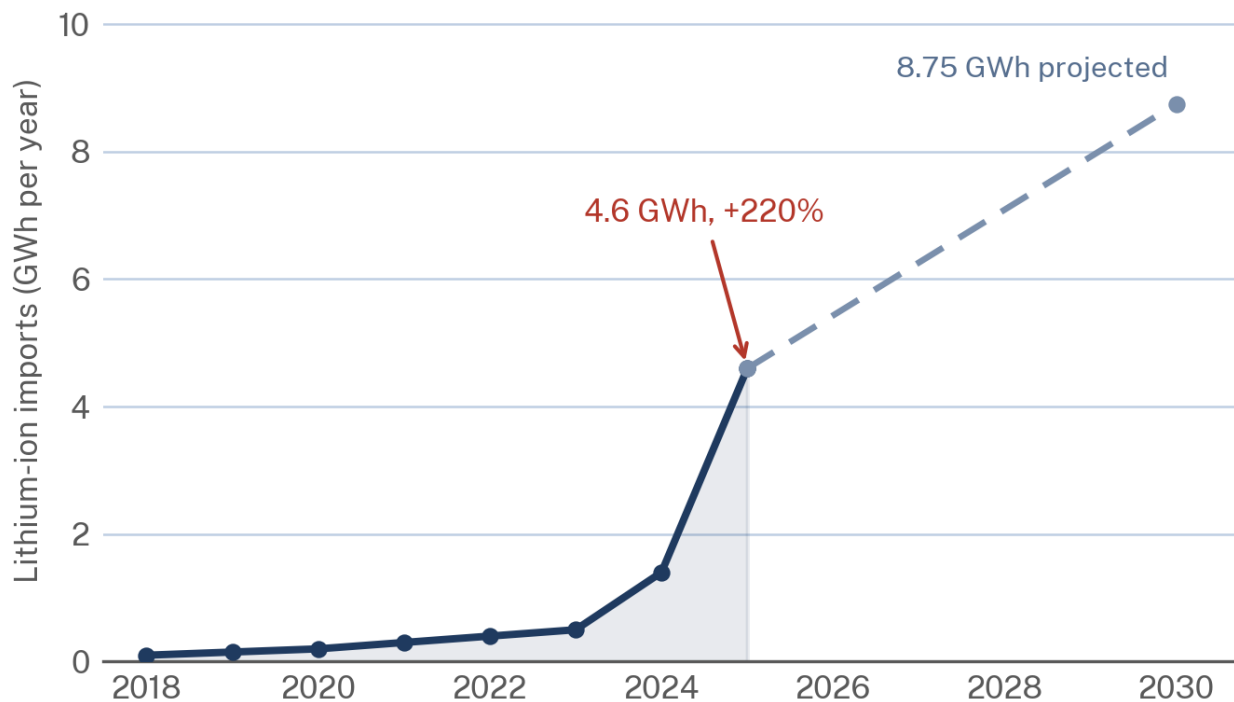


Figure 12. About 60% of everything imported since 2018 arrived in a single calendar year

Annual lithium-ion battery imports, Pakistan, GWh, 2018 to 2025, with the business-as-usual projection to 2030

Source: ARS analysis of Renewables First¹ for 2018 to 2025 and IEEFA⁴² for the 2030 projection.

Notes: The dashed segment is a business-as-usual projection, not a reported figure. The Renewables First series counts storage systems; IEEFA's 2024 pack import figure is 1.25 GWh.

Annual imports ran between one tenth and half a GWh from 2018 through 2023, essentially a niche. In 2024 that jumped to 1.4 GWh. In 2025 it more than tripled to 4.6 GWh, a 220% year on year increase.⁴³ Early indications suggest the pattern held through the first half of 2026, with March and April running particularly high.

A separate projection from the Institute for Energy Economics and Financial Analysis put 2024 battery pack imports specifically at approximately 1.25 GWh, with a further 400 MWh in the first two months of 2025, and projected imports reaching 8.75 GWh by 2030 on a business as usual basis.⁴²

The two series are close but not identical, and the difference is one of definition rather than disagreement. Renewables First counts imports of lithium-ion battery energy storage systems, 1.4 GWh in 2024; IEEFA counts battery pack imports, 1.25 GWh in the same year.^{42,43} This report adopts the Renewables First series for 2018 to 2025 because it is the only continuous annual series available and because it captures complete systems, the market a local assembler would

first address. The IEEFA figures are used only for the 2030 projection, which Renewables First does not provide.

The growth in battery imports is not simply a demand signal the government wants to accelerate without qualification. It is a trend Pakistan's own energy regulators are simultaneously trying to manage, given knock on effects for a power sector already carrying circular debt and stranded capacity obligations from earlier generation investment. Anyone assuming that policy will move unambiguously toward maximising battery adoption is missing a genuine internal tension within the government's own position, and that tension is likeliest to surface as pressure on exactly the charging tariff discount discussed in section 4.3.

This import growth sits directly alongside Pakistan's rooftop solar boom, which is its principal driver. Distributed solar capacity reached an estimated 38 GW by June 2025, 27 GW of it installed in the preceding two years, one of the fastest expansions of distributed solar anywhere in the world over that period.² Official statistics capture only the net metered fraction, 4.9 GW by March 2025,³⁶ so the estimate is triangulated: Chinese export data showing 47 GW of panels shipped to Pakistan set an upper bound, satellite imagery gives 27.5 GW, and a household survey projects 33 GW. The figure should be read as a point within that range rather than a census. The Federal Minister for Power has unveiled a roadmap targeting 90% clean electricity by 2035, up from roughly 55% currently,⁴⁴ a target implicitly assuming continued rapid growth in both solar and the storage needed to make it achievable. Importing the hardware behind that transition indefinitely would represent both a missed opportunity and a growing balance of payments exposure.

The pricing data attached to this surge explains why local manufacturing would represent a genuine cost advantage rather than merely a nationalist preference. IEEFA estimates that taxes and customs duties add 48% to the cost of imported batteries,⁴² a premium that reflects fiscal policy rather than any underlying difference in global cell costs. This produces a genuine policy paradox. The tariff structure intended to protect and incentivise future domestic manufacturing is, in the interim, sustaining elevated consumer prices in a manner that retards the adoption curve the wider policy seeks to accelerate. Resolving that tension constitutes one of the substantive design challenges facing the framework discussed in section 3, and it admits of no obvious solution, since the two objectives operate in direct opposition over the short run.

For anyone building an import substitution thesis, the disciplined approach is to disaggregate rather than treat battery imports as one line. Finished packs and complete storage systems are where local assembly can capture value in the near term, closest to demonstrated capability. Modules sit one step up. Cells, cathode active material, anode material and electrolyte and separator components are progressively harder, requiring process engineering capability Pakistan does not currently possess anywhere in its industrial base. A thesis built on pack and module substitution is grounded in demonstrated capability against a large and rapidly growing addressable volume. A thesis built on cell or cathode substitution is a fundamentally different proposition requiring capability developed from a standing start, and folding the two into one growth narrative is precisely the error this report exists to correct.

5.3 The Mineral Base: Geology Against Battery-Grade Reality

Pakistani officials have repeatedly cited domestic phosphate, iron and manganese deposits as the rationale for prioritising the lithium iron phosphate battery type, on the logic that Pakistan possesses a real base for the two elements giving the battery technology its name in a way it does not for nickel or cobalt. The claim deserves to be taken seriously rather than dismissed, because at the level of geological occurrence it is substantially true. It also deserves considerably more scrutiny than it has received, because geological occurrence and battery-grade processed material are separated by an enormous amount of metallurgical capability that Pakistan has not demonstrated at scale.

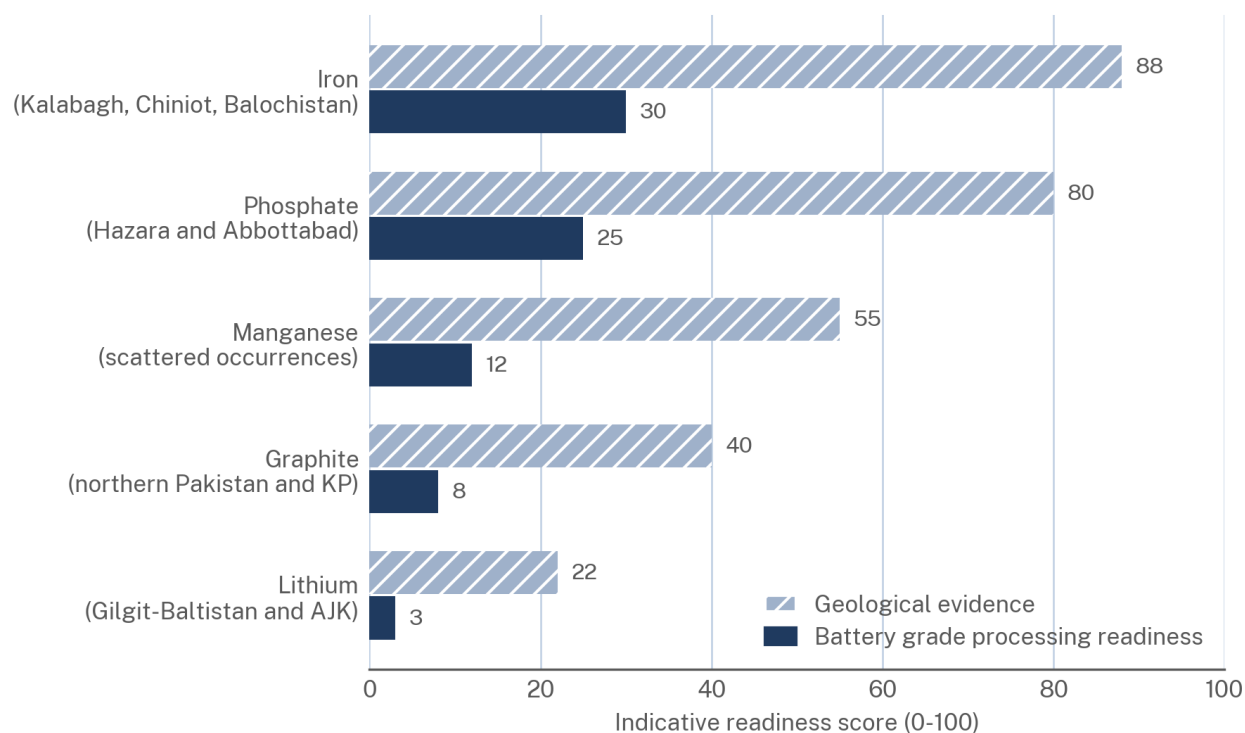


Figure 13. The rock is real; the processing capability that turns rock into battery-grade material largely is not

Geological evidence against battery-grade processing readiness by input, ARS analytical scoring, 2026

Source: ARS analysis of Geological Survey of Pakistan material,^{45,46} Punjab Mineral Company⁴⁷ and United States Geological Survey data.⁴⁸

Notes: Analytical scoring based on documented survey evidence, not proven reserve tonnage. Scores are ordinal and rank the inputs against one another.

Working through the inputs individually, iron ore is documented at several sites: the Kalabagh deposit in Mianwali district of Punjab,⁴⁷ the Chiniot project in Punjab and the Nokkundi and Chagai deposits in Balochistan,⁴⁸ although the Geological Survey of Pakistan has long noted that most deposits located are either small or low in grade.⁴⁵ Phosphate is established as a mining category, with an economic base in the Hazara division around Abbottabad and newer occurrences identified in recent surveys whose grades of 12% to 25% phosphorus pentoxide are described as suitable for the fertiliser industry.⁴⁶ Lithium potential in the northern mountain belt has been noted by the Geological Survey,⁴⁵ but there is no proven commercial supply anywhere. The survey literature reviewed for this report gives little consistent information on manganese or graphite, the latter relevant to the anode rather than the cathode.

The distinction running through all five is between a documented occurrence and proven commercial viability. An iron ore deposit entirely suitable for conventional steelmaking may require a different and considerably more expensive and complex purification process to reach battery-grade cathode purity, where trace impurities acceptable in structural steel degrade cell performance and safety. The same applies to phosphate, where the Hazara and Abbottabad base

has been developed and optimised for fertiliser grade output over decades. Converting that supply chain, or building a parallel one, to battery-grade purity is a substantial engineering and capital undertaking that has not, on the evidence available, begun at meaningful scale.

For lithium the gap is starker still. Exploration stage resources by definition have not been proven at a scale or grade supporting commercial extraction, let alone battery-grade refining. Pakistan should not be counted as having a domestic lithium supply in any near term sense regardless of how the mineral base narrative is framed in policy documents, and an investor who reads the official framing as implying otherwise is being misled by omission rather than assertion.

This matters enormously for how the entire localisation strategy should be read. The rationale for choosing this battery technology rests on the claim that Pakistan has the raw material base to support it domestically. That claim is true at the geological level for iron and phosphate, weaker for manganese, and essentially untrue for lithium and graphite in any near term commercial sense. A genuinely domestic supply chain sourcing battery-grade iron and phosphate from Pakistani mines remains a plausible medium to long term outcome given the underlying geology, but it is not a current reality, and no public investment material or survey document this report could access establishes that battery-grade processing of Pakistani iron or phosphate has been proven, let alone scaled. This is precisely the kind of open item a Chinese investor should insist on verifying independently through a specific geological and metallurgical assessment against battery-grade specifications, rather than accepting the general narrative from ministerial statements.

There is a comparative lesson here that Pakistani policymakers and the investors evaluating their claims would do well to keep in view. Indonesia's nickel processing buildout, frequently cited as the template for how a resource endowed developing country forces its way up the battery chain and examined in section 7.3, took the better part of a decade,⁴⁹ required tens of billions of dollars in smelting and refining capital supplied predominantly by Chinese partners rather than domestic capital, and depended on an ore export ban that left foreign processors no alternative but to build onshore. Pakistan has neither the scale of proven reserves represented by Indonesia's nickel base, nor an equivalent export restriction mechanism, nor committed capital approaching what that buildout requires. The underlying iron and phosphate geology is real and relevant to this specific type of battery in a way few developing markets can claim. The timeline for converting it into a functioning battery-grade supply chain is

simply far longer and more capital intensive than the current policy narrative suggests.

5.4 Power, Water and the Cost of Making Cells

The staged path this report recommends ends in cell manufacturing and battery-grade material processing, and both are among the most power-intensive activities in the value chain. A peer reviewed study of current cell production finds that making a cell with one kilowatt hour of storage capacity consumes between 20 and 40 kWh of energy, much of it in electrode drying, dry room climate control and cell formation.⁵⁰ A plant producing 5 GWh of cells a year would therefore draw in the order of 100 to 200 GWh a year, the consumption of a large industrial estate, and would need that supply without interruption, since dry rooms and formation cycles cannot be paused without scrapping work in progress.

Price is the first obstacle. At about 13.5 US cents per kWh, Pakistan's average industrial rate is more than twice the rate in India and well above China's 7.7 cents, on International Energy Agency data.⁵¹ In October 2025 the government cut the rate on incremental industrial consumption from Rs. 34 to Rs. 22.98 per kWh for three years, roughly 8 US cents at current exchange rates and close to China's average.⁵² A new plant's load would appear to be incremental in full, but a cell or cathode project should seek written confirmation that its load qualifies, and treatment beyond the three year window, since the payback period on a cell line is considerably longer.

Reliability is the second. The grid problem Pakistan faces is not a generation shortage; the Power Minister has said that the challenge has shifted from generation to storage.⁵³ The weakness lies in distribution. Transmission and distribution losses and recovery shortfalls at the distribution companies cost Rs. 265 billion and Rs. 132 billion respectively,⁵⁴ and the feeders that serve industrial estates share in that weakness. A cell plant should budget for a dedicated high-voltage connection, on site storage and backup generation as part of its capital cost rather than treat grid supply as given.

Water is the third, and the least examined. Cathode material production and cell manufacturing use substantial volumes of process water, and the phosphate and iron processing on which the lithium iron phosphate case depends adds more. None of the policy documents reviewed for this report addresses water supply for battery manufacturing, and Karachi's industrial zones, the most likely location given port access, already face chronic water shortages. A water supply assessment should sit alongside the power assessment in any feasibility study.

None of this rules out the lower stages. Pack assembly, battery management system integration and testing draw modest power and can run on the solar and storage supply that Pakistani industry is already installing. The power constraint bites at the cell and material stages, which reinforces the case for staging set out in section 5.1.

5.5 Workforce and Technical Depth

Pakistan does not lack engineers. The Pakistan Engineering Council has reported some 260,000 registered engineers and, in 2021, estimated that more than 50,000 were without work.⁵⁵ At the assembly end the base is broad and growing: the government had licensed 57 electric vehicle manufacturers by February 2025, 55 of them for two and three wheelers,⁵⁶ and the two wheeler industry has decades of experience of high volume production with over 90% local content.

The gap is specialised. Battery management system design, cell process engineering, electrochemical quality control and formation management are disciplines for which this report could identify no established training pipeline in Pakistan, and no public dataset measures how many Pakistani engineers hold those skills. The regional experience is instructive. Chinese battery firms in Thailand assembled packs from imported cells or supplied production technology before Sunwoda became the first to be approved for cell production there in March 2025,⁵⁷ and Indonesia's first cell plant was built and run by Hyundai and LG Energy Solution.⁵⁸ In both cases the technology owner supplied the process knowledge and the early technical staff.

For an investor, the implication is that pack and battery management system operations can be staffed largely from the existing engineering pool after a period of training, while any cell line should budget for expatriate process engineers in its first years and a structured transfer programme, written into the technology licence, to replace those engineers over time. A training commitment of that kind is also the most credible local content an early cell project can offer, and it is worth negotiating as a counterpart to incentives.

6 The Chinese Investment Tracker

Six named Chinese engagements in Pakistan's electric vehicle and battery sector could be identified from the public record at the date of this report. They differ enormously in what they actually commit, and this section sorts them by that criterion rather than by the prominence of the announcement.

6.1 Named Engagements, Sorted by What They Actually Commit

The most advanced and best documented Chinese linked investment in the sector is the joint venture between Hub Power Holdings Limited, a wholly owned subsidiary of HUBCO, Pakistan's largest independent power producer, and Mega Conglomerate Private Limited, itself HUBCO's largest single shareholder at roughly 19.5%. The vehicle is Mega Motor Company Private Limited, split exactly 50:50 between the two partners.⁵⁹

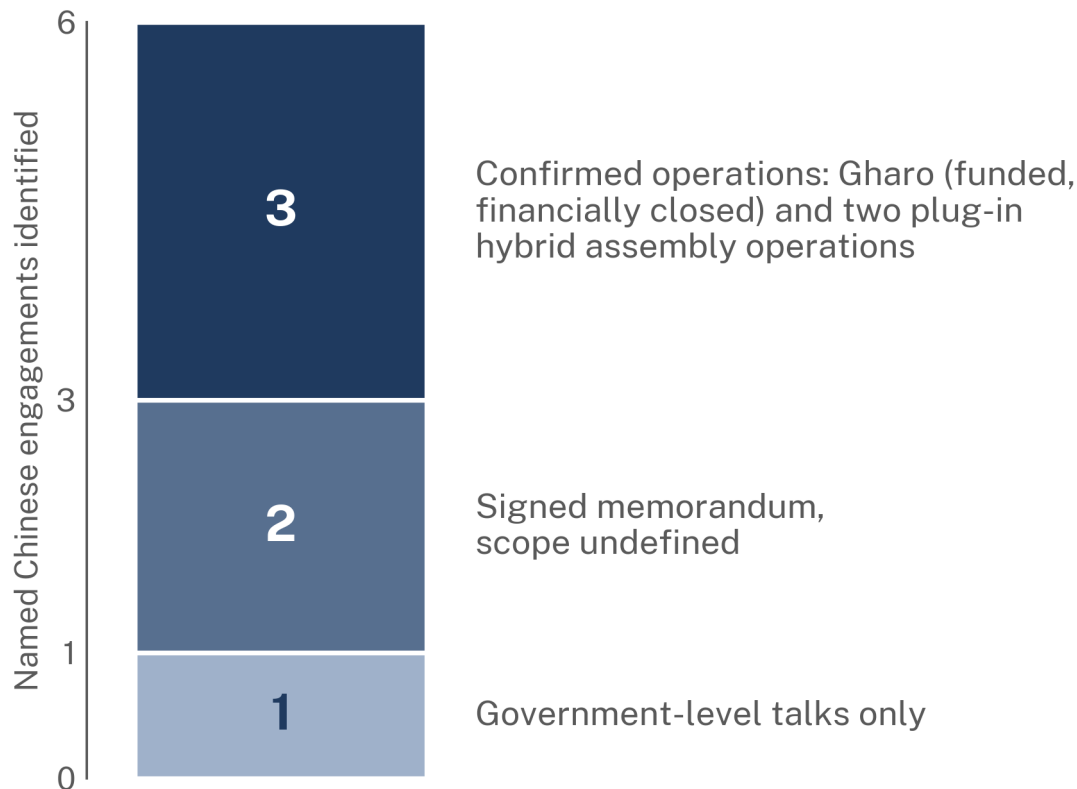


Figure 14. Three of the six named Chinese engagements are confirmed operations; the rest are memoranda or talks

Composition of named Chinese engagements in Pakistan's EV and battery sector, by commitment level, September 2026

Source: ARS analysis of the engagements listed in Appendix B.

Notes: Counts reflect engagements identifiable from the public record at the date of this report and are not a complete census. Jetour's petrol-only assembly is outside scope and not counted.

BYD's actual role in that structure is frequently misreported and is worth stating precisely. BYD has not taken an equity stake in Mega Motor Company. Its involvement runs through a master supply and manufacture agreement, a technical licence agreement and a distribution arrangement, giving it control over technology, branding and component supply without carrying direct equity or balance sheet exposure to the Pakistani venture. This is a common structure for Chinese automakers entering markets where they want a manufacturing foothold without committing capital directly, and understanding it matters for

locating where the financial risk actually sits, which is overwhelmingly with the two Pakistani partners rather than with BYD.

The project is a CKD assembly plant at Gharo in Sindh, with a reported cost of approximately USD 150 million on a 60:40 debt to equity basis, designed for 25,000 vehicles a year.^{60,61} Financial close was achieved in January 2026, with commercial operations then expected in the second half of 2026. HUBCO describes the venture as its entry, through Hub Power Holdings and Mega Motor Company, into electric vehicles with BYD from June 2024.⁶² Both foreign and local lenders participated in the debt financing with a term sheet signed ahead of close, indicating conventional project finance discipline and independent lender diligence rather than a purely strategic commitment. The credit rating agency assessing HUBCO has separately noted that the planned electric vehicle investments are expected to materially increase group leverage, which is a candid acknowledgment that this is a genuine balance sheet commitment rather than a costless option.⁵⁹

The plant's timetable has since slipped. Reporting in September 2026 records that the facility, originally scheduled to become operational in the first half of financial year 2026, missed that deadline, with Hub Power Company informing analysts that the launch would fall in the second half of calendar 2026.⁶³ The same reporting puts the project financing component at USD 90 million within the USD 150 million total, capacity at 25,000 units scalable to 50,000, and a stated target of 30% share of the combined electric and plug-in hybrid segment by 2030. No explanation for the delay was provided to the reporting journalist. The slip-page does not alter the classification of the project but it does bear directly on the first of the triggers set out in section 9.

What makes this a useful test case is that it is, by its own description, assembly rather than manufacturing. 25,000 units a year is meaningful for the Pakistani market, roughly a month or two of total national passenger vehicle sales at current run rates, but volume says nothing about how much value is created domestically. The real test will come when the Gharo line begins commercial production and the actual local content of vehicles rolling off it can be assessed against whatever requirements are in force at that point.

On the battery side, Treet Battery Limited signed a memorandum of understanding on 25 May 2026 with Contemporary Nebula Technology Energy Company (CNTE), in which CATL holds a strategic investment.⁶⁴ This should be understood exactly for what it is. It is a real, publicly disclosed agreement between two named entities, establishing a framework to explore energy storage cooperation

and potential introduction of CATL cell based products into Pakistan, leveraging Treet's existing distribution and after sales network. It is not a commitment to cell production, and it should not be classified alongside the Gharo project as manufacturing investment. It is a confirmed commercial relationship and a credible pathway toward CATL linked technology, but its own language stops well short of committing either party to local assembly.

Direct engagement between the Government of Pakistan and CATL itself sits earlier still. On 24 May 2026 CATL's Executive President Oscar Luo met Prime Minister Shehbaz Sharif in Hangzhou,⁶⁵ with discussions covering advanced battery technology, storage solutions and investment across renewable energy and electric mobility. Pakistan's ambassador to Beijing separately confirmed active discussions, framing them within the second phase of the China Pakistan Economic Corridor, which has shifted emphasis from power generation and infrastructure projects towards manufacturing and technology transfer. No public reporting discloses an investment figure, plant capacity or implementation timetable. This is an engagement at the talks stage, significant as a signal of top level interest from the world's largest battery manufacturer, but categorically different from either the Gharo project or the Treet memorandum.

Completing the tracker, a memorandum between StarCharge Energy Pakistan and Bahum Global announced in July 2026 covers charging ecosystem development alongside local manufacturing and technology transfer ambitions, with scope and timeline still to be defined.⁶⁶ On the vehicle side beyond BYD, two Chery-linked operations are assembling plug-in hybrids. Nishat's NexGen Auto began local production of the Jaecoo J7 plug-in hybrid in October 2025 at a new Faisalabad plant built with a USD 100 million investment,⁶⁷ and Master Group's Chery venture launched a locally assembled Tiggo 8 plug-in hybrid in January 2026, followed by the Tiggo 9 plug-in hybrid, whose assembly status the public record does not establish.⁶⁸ Jetour's local assembly through United Motors is of turbocharged petrol models⁶⁹ and falls outside the scope of this report, so it is not counted. Each counted operation should be seen through the same lens applied to Gharo: genuine and welcome investment in assembly capacity, not manufacturing in the real sense.

6.2 Structuring an Entry Against What This Evidence Shows

For an investor evaluating fresh entry rather than observing existing deals, the accumulated evidence points toward several structuring principles beyond the staged approach already set out in section 3. The first is to separate the equity structure from the technology relationship, following the precedent BYD itself

established. Providing technology, branding and component supply through licensing and supply agreements rather than direct equity allows participation in Pakistani market growth while limiting balance sheet exposure to currency volatility, an unsettled tariff framework and the broader macroeconomic and political risk examined in section 6.3.

That is not a universal recommendation. A manufacturer with genuine ambitions to build deep local capability rather than simply access the market through a licensed partner may reasonably prefer direct equity to maintain control over execution and quality. But for an investor whose primary interest is market access and brand presence, the licensing structure deserves serious consideration as the lower risk entry, and the fact that the most sophisticated Chinese entrant to date chose exactly this route is evidence worth weighing rather than coincidence.

The second principle is to insist on defining measurable technical milestones that separate the value chain stages set out in section 5, rather than leaving them as loosely worded aspirations. Both the Treet and StarCharge agreements are currently structured as frameworks for exploring cooperation rather than binding commitments with defined deliverables. That is a reasonable first step in an unfamiliar market. But any investor entering a similar framework should push, in subsequent definitive agreements, for specific contractually defined indicators around cell, module and pack level output that allow both parties, and any outside observer conducting later diligence, to assess objectively whether the partnership is progressing up the ladder or remaining static at assembly and distribution indefinitely.

The third concerns how to treat unfinalised policy when negotiating today. Rather than waiting for finalisation, which risks losing first mover positioning, or signing binding long term commitments premised on pitchbook figures as though they were law, the defensible middle path builds explicit renegotiation triggers into definitive agreements, specifying that key terms around tariff treatment, local content obligations and incentive dependent capital commitments will be revisited on formal gazettal if notified terms differ materially from what circulated during negotiation. This is standard practice for industrial investment in jurisdictions with genuinely evolving frameworks, and its absence from an agreement signed in Pakistan today should be read as a signal about the drafting party's sophistication rather than as confidence in the policy trajectory.

The fourth follows from the PAVE record in section 3.2 and applies specifically to any project whose economics depend on a government disbursement rather

than a tariff concession. Tariff concessions operate at the border and require no administrative discretion once notified. Subsidy disbursements require a functioning payment chain, and Pakistan's demonstrated record on that chain is 13% delivery in a pilot whose budget had been allocated. Any project relying on subsidy flow rather than duty relief should model a substantial delay assumption and, where possible, negotiate for the tariff based version of an incentive over the disbursement based version even at a lower headline value, because a smaller concession that operates automatically is worth more than a larger one that depends on an administrative process with this track record.

Finally, given the distinction between CATL as a corporate entity and the separate company in which it holds a strategic investment, any investor should be precise in how it characterises this relationship in its communications and internal risk assessments. Describing a Treet style partnership as CATL entering Pakistan could be misleading. The counterparty is a separate CATL backed company operating under a memorandum, while CATL itself remains at the stage of preliminary government discussions. This kind of mischaracterisation can erode credibility with sophisticated counterparties as soon as the underlying facts are examined.

6.3 Currency, Balance of Payments and Repatriation Risk

Currency and external payment risk deserve direct treatment, because in 2022 and 2023 they did more damage to Pakistan's vehicle assemblers than any change in industrial policy. With reserves falling, the State Bank of Pakistan restricted imports, and completely knocked down kits were among the goods affected. Pak Suzuki shut its car plant repeatedly in late 2022 and January 2023 for lack of kits, citing State Bank restrictions on imports under HS code 8703.⁷⁰

Repatriation was restricted at the same time. Profit and dividend repatriation by foreign investors fell from USD 1.68 billion in financial year 2022 to USD 331 million in financial year 2023, an 18 year low.⁷¹ The backlog was cleared in 2024 once reserves recovered, and repatriation reached USD 2.21 billion in financial year 2024.⁷² Both halves of the episode matter to an investor: restrictions were imposed abruptly when reserves ran short, and they were lifted only once an International Monetary Fund programme had restored reserves.

That programme remains the anchor. Pakistan's 37 month Extended Fund Facility of about USD 7 billion was approved in September 2024,⁷³ and its third review was completed in May 2026.⁷⁴ The programme stabilises the external position but also constrains industrial policy. It is the context for the tariff rationalisation discussed in section 3.4, and the Power Minister has said that commitments

under the programme have held back tariff reforms that could lower electricity costs.⁷⁵

Three consequences follow for structuring. First, a project that imports cells, kits or equipment should treat access to foreign exchange for imports, not only the exchange rate, as a risk to be priced; supplier credit from the Chinese technology partner and inventory buffers are the practical hedges. Second, a project that sells in rupees and services foreign currency debt carries the full exchange rate risk, which argues for local currency financing where it is available and for export revenue where the product allows. Third, the repatriation record argues for structuring part of the return as licence fees and supply margins earned by the Chinese partner outside Pakistan, which is a further reason the BYD licensing model examined in section 6.1 deserves weight. None of these measures removes the risk, and a project whose economics fail under a repeat of 2022 and 2023 should not proceed on the assumption that the episode will not recur.

7 Pakistan Against Its Regional Competitors

Pakistan does not compete for Chinese electric vehicle and battery capital in isolation. Four comparators are examined, selected against the rule stated in section 2.2: Bangladesh as the closest South Asian peer, Vietnam as the established export manufacturing destination, Indonesia as the resource-led case and Thailand as the regional centre of Chinese electric vehicle assembly. Each is pursuing the same investors with a materially different offer, and the comparison clarifies what Pakistan is actually selling.

7.1 Bangladesh and the Discipline of a Sunset Clause

Bangladesh's approach, set out in its financial year 2026-27 budget proposals, is built around a design principle Pakistan's framework has not matched in clarity, even though both countries are pursuing broadly similar goals.⁷⁶ The core mechanic is a duty structure tiered explicitly by the degree of local value addition a manufacturer actually undertakes, rather than a flat concession available regardless of contribution. Manufacturers carrying out body fabrication, welding, painting and assembly of three and four wheeled electric vehicles with high local value addition receive exemption from essentially all duties and taxes on imported materials and parts, retaining only a nominal 3% import duty. Manufacturers conducting lower value operations, essentially assembly and painting without fuller fabrication, face 15% on the same inputs.

That five fold differential between 3% and 15% creates a sharper and more legible incentive to move up the value chain than anything currently notified in

Pakistan, where the completely knocked down tier system pushes in the same direction without as clean a binary. It is worth noting that Bangladesh achieves this within a single budget instrument, where Pakistan's equivalent is spread across an expired auto policy running on extension, an unfinished battery policy and a separate unfinished battery specific instrument.

On the consumer side, Bangladesh is cutting the total tax burden on imported electric vehicles priced up to USD 25,000 from 93% to 64%, with vehicles up to USD 50,000 facing 80%.⁷⁷ The advance income tax on registration, previously a flat and often prohibitive BDT 200,000 regardless of specification, is being re-structured to scale with battery capacity, falling to BDT 25,000 for 200 kW vehicles and capped at BDT 75,000 for the largest category. Charging infrastructure receives dedicated treatment, with the previously applicable 39.75% duty and tax burden on imported chargers eliminated entirely, a more aggressive charging incentive than anything currently notified in Pakistan. Simultaneously the tax burden on petrol and diesel cars between 1,200cc and 1,600cc rises from 132% to 156%, pushing from both directions at once.

The feature that most clearly distinguishes the Bangladeshi design, and the one carrying the greatest instructive value, is the explicit sunset attaching to concessions on the most strategically significant components. Duty and tax exemptions on imported raw materials and components for lithium-ion and sodium-ion battery production run to 30 June 2030, while those on mounting structures, lithium cells, battery packs and energy storage systems are to be withdrawn after June 2028, with the stated expectation that local production will meet demand by then.⁷⁸ This gives established players a bounded window to establish functional cell and pack production before importing the same components becomes substantially more expensive. The structure is considerably clearer than an open ended assembly subsidy, and it stands in marked contrast to developing country incentive schemes that have remained in force for decades beyond the point at which they served their original purpose.

POLICY RECOMMENDATION

The Bangladeshi sunset clause is the single most transferable design feature identified in this comparison. A bounded window on component concessions converts an open-ended assembly subsidy into a schedule that forces the localisation the policy exists to produce, and Pakistan's framework currently contains no equivalent mechanism.

It would nonetheless be a mistake to read this more codified structure as evidence that the underlying market is ahead of Pakistan. Bangladesh Road Transport Authority figures record only 669 electric vehicles formally registered nationally as of 14 May 2026, with formal registration having begun only in September 2022.⁷⁹ That is a very small base, and it means Bangladesh's well designed architecture is a bet on future growth rather than a reflection of a thriving present. Some Bangladeshi business leaders have themselves warned that the incentive structure may inadvertently favour imports over domestic manufacturing in early implementation, which echoes precisely the assembly against manufacturing distinction this report applies to Pakistan and confirms that a better designed policy on paper does not automatically produce a better outcome in practice.

For a Chinese investor weighing the two, the honest comparison is between a market with a smaller current base but a more legally certain and explicitly tiered structure with a defined sunset, and a market with a comparably small base but a considerably larger population, a longer established assembly tradition through decades of Japanese manufacturer operations, and a framework that remains substantially less codified. Neither offers a mature ecosystem. The choice comes down to which risk an investor is better positioned to manage: policy uncertainty in Pakistan or a genuinely thin current market in Bangladesh, rather than any objective superiority of one over the other.

7.2 Vietnam and the Value of a Codified Framework

Vietnam belongs to a different category of comparison, not because its individual incentives are necessarily more generous, but because it enters with an established domestic manufacturing base and a legal architecture for foreign investment incentives comprehensively overhauled within the past year. The National Assembly passed an entirely new Corporate Income Tax Law on 14 June 2025, replacing the previous framework in full, effective from 1 October 2025 and applicable retroactively to the whole 2025 tax year.^{80,81} This is a complete legal overhaul rather than incremental amendment, and its recency means an investor evaluating Vietnam today works from the current architecture rather than an older framework layered with successive patches.

The regime follows a familiar tax holiday structure applied through more tightly defined criteria than before. Qualifying projects, including high technology application, strategic technology and supporting industry categories that a genuine cell manufacturing or advanced material processing facility would plausibly meet, may receive a full corporate income tax exemption following the first

profitable year of activity, followed by a period at half the standard rate, with the duration varying by category and location. The preferential rates most commonly applied are 10% for 15 years and 17% for 10 years, and projects licensed before the new law keep their existing incentives,⁸² which gives a sense of the order of magnitude involved.

A significant structural change removes what had been an easy qualifying route. Locating within an ordinary industrial zone no longer qualifies a project for incentives on its own.⁸³ Projects must now qualify through genuine sector criteria demonstrating high technology, digital or green characteristics, or through geographic criteria specific to special economic zones. Projects licensed before the new law may continue under previous terms until expiry. This tightening reflects Vietnam's own assessment that blanket industrial zone incentives had become insufficiently targeted, and it is worth noting as evidence of a state actively refining its incentive architecture rather than simply accumulating concessions.

Beyond corporate tax, Vietnam layers additional measures the South Asian markets do not match in combination. Decree 202 of 2026 extends full exemption from the registration fee for battery electric vehicles through 31 December 2030, framed explicitly by Vietnamese officials as providing the policy certainty manufacturers require when committing capital on the 5-10 year horizons typical of automotive decisions, which is precisely what this report has identified as missing in Pakistan.⁸⁴ Qualifying research and development expenditure has been deductible at 200% of actual cost since 1 October 2025,⁸⁵ a genuinely generous innovation incentive exceeding anything on offer in either South Asian market.

One complication a sophisticated investor must weigh has no equivalent in the Pakistani or Bangladeshi context. Vietnam has implemented the global minimum tax framework through a domestic decree effective from mid October 2025 with retroactive application to financial year 2024, applying to multinational groups with consolidated global revenue of at least EUR 750 million in two of the preceding four years.⁸⁶ Both CATL and BYD comfortably clear that threshold. For groups of that scale, the 15% effective minimum enforced at consolidated level can claw back much of the benefit of a zero or low preferential rate applied at the Vietnamese subsidiary level. This is technical but material, and it means Vietnam's headline incentive package is worth considerably less to exactly the largest Chinese battery manufacturers most likely to be considering an investment than it is to smaller investors falling below the threshold.

That asymmetry is worth stating plainly because it partially closes a gap that headline comparison would suggest is unbridgeable. If Vietnam's principal

advantage over Pakistan were purely the generosity of its tax holidays, the comparison would be straightforward. Because a substantial portion of that generosity is neutralised by the global minimum tax for the specific investors most relevant to this sector, Vietnam's real remaining advantages are its existing ecosystem, its infrastructure and its policy predictability, rather than its net fiscal offer. Those are real advantages, but they are a narrower set than the raw incentive comparison implies.

7.3 Indonesia and the Limits of a Resource-Led Strategy

Indonesia is the case Pakistani officials invoke when they argue that mineral endowment can be converted into battery manufacturing, and it is the right benchmark for the claim examined in section 5.3. The central instrument was a ban on exports of unprocessed nickel ore, first imposed in 2014 and reinstated from 1 January 2020. Chinese firms became the dominant investors in the processing capacity that followed, and the value of Indonesia's nickel product exports rose from less than USD 1 billion in 2015 to nearly USD 20 billion in 2022.⁴⁹

Cell manufacturing followed more slowly. Indonesia's first battery cell plant, a joint venture of Hyundai Motor Group and LG Energy Solution with 10 GWh of annual capacity, opened in Karawang in July 2024, four and a half years after the ban took effect.⁵⁸ The largest integrated project, a USD 6 billion chain from nickel mining to cathode, cells and recycling led by a CATL subsidiary with the state partners Antam and Indonesia Battery Corporation, broke ground in June 2025 with a first phase cell capacity of 6.9 GWh.⁸⁷ The path has not been smooth: an LG-led consortium left a separate project of over USD 7 billion in 2025, citing market conditions and trade tensions, and Huayou Cobalt was reported to be taking its place.⁸⁸

On the vehicle side, Indonesia paired the resource strategy with demand incentives tied to local production. Value added tax on electric vehicles meeting a 40% local content requirement was cut from 11% to 1%, and importers of completely built electric vehicles were exempted from import duty and luxury tax on condition that they build local plants by 2026, backed by bank guarantees equal to the duties waived.⁸⁹ BYD's Subang plant, with capacity of 150,000 vehicles a year, was inaugurated in September 2026.⁹⁰ Indonesia applies the global minimum tax from 2025,⁹¹ so its income tax holidays are worth less to the largest Chinese groups, while its value added tax and import concessions are not affected.

The Indonesian lesson for Pakistan is mixed. The export ban worked because Indonesia holds the world's largest nickel reserves,⁹² which gave it the leverage to force processing onshore; Pakistan's iron and phosphate base confers no

comparable leverage, because neither mineral is scarce globally. What does transfer is the design of the demand incentives: Indonesia made its consumer and import concessions conditional on dated local production commitments backed by guarantees, a mechanism Pakistan's framework does not use. What also transfers is the timescale. Even with decisive leverage and billions of dollars of committed capital, the first cell plant took four and a half years from the ban to open.

7.4 Thailand and the Assembly Hub Model

Thailand is where Chinese electric vehicle makers have actually built capacity in the region, and it tests the proposition that assembly incentives lead to deeper manufacturing. BYD opened its Rayong plant in July 2024, a USD 490 million investment with capacity of 150,000 vehicles a year, and said that it would also assemble batteries and other key parts there.⁹³ Electric vehicles reached 17.7% of Thai car sales in the first seven months of 2025, up from 11.4% in 2024.⁹⁴

The instrument behind this is the EV 3.5 scheme. It combines a purchase subsidy of up to THB 100,000 in 2024 and THB 75,000 in 2025 for eligible cars, a 2% excise rate on battery electric vehicles and import duty relief for 2024 and 2025, and it requires manufacturers that took the import relief to build two vehicles in Thailand for every one imported from 2026, and three for every one from 2027.⁹⁵ Battery cell projects can receive corporate income tax holidays of up to eight years from the Board of Investment, though Thailand's top-up tax, in force from 2025, limits their value to the largest groups.⁹⁶ The production offset is the distinctive feature: it converts an import incentive into a dated obligation to manufacture locally, a harder commitment device than either Pakistan's localisation schedule or Bangladesh's sunset clause.

Thailand also shows the limits of the model. Chinese battery firms including Gotion, Svolt, CATL and Eve Energy supplied packs assembled from imported cells, or production technology, before Sunwoda's approval in March 2025 made it the first to produce cells in Thailand, some years after the vehicle plants arrived.⁵⁷ In May 2026 ten Thai automotive associations warned that domestic production would struggle to compete with Chinese imports once incentives end in 2027, and proposed import quotas tied to local production and stricter enforcement of a 40% local content threshold.⁹⁷ The Thai experience confirms the central distinction of this report: a large, well incentivised assembly base does not produce cell manufacturing on its own, and the step from one to the other has to be designed and paid for separately.

7.5 What the Comparison Actually Resolves

Taken together, the choice facing an investor is not a ranking exercise yielding a single winner but a genuine trade-off between differing combinations of certainty, maturity and unrealised upside. Figure 15 sets out this report's assessment across six dimensions.

	Market size and upside	Existing ecosystem	Policy certainty	Chinese capital committed	Charging infrastructure	Net incentive value
Pakistan	9	3	2	5	2	4
Bangladesh	7	2	7	1	1	8
Vietnam	6	8	9	7	7	5
Indonesia	9	9	5	10	5	6
Thailand	5	8	7	9	6	7

Figure 15. Pakistan matches the largest comparator on upside but trails all four on policy certainty

Comparative scorecard across six dimensions, Pakistan and four comparators, ARS analytical scoring, 2026

Source: ARS analysis of the evidence reviewed throughout this report.

Notes: Analytical scorecard, not an official ranking. Scores run from 0 to 10, are ordinal rather than cardinal, and are not summed. Scale anchors and the evidence behind each score are in Appendix C.

Vietnam offers the most legally certain framework, backed by a functioning domestic manufacturing base and a freshly overhauled tax law, but bundled with a more crowded competitive landscape, higher operating costs than either South Asian alternative, and for the largest investors the global minimum tax offset just discussed. Bangladesh offers the most cleverly tiered incentive design of the five, with a defined transition timeline forcing genuine localisation on a schedule rather than allowing indefinite import dependent assembly, but sitting on a market so early stage that any case must rest almost entirely on projection.

Indonesia offers the deepest upstream integration and the most assertive state, with a cell plant operating and a mining to recycling chain under construction, but its leverage rests on nickel, which gives it no advantage in the lithium iron

phosphate chemistry Pakistan has chosen. Thailand offers the largest installed base of Chinese vehicle assembly and a dated production obligation that makes its incentives self-enforcing, but a crowded field and a domestic industry already warning about competition from imports.

Pakistan offers one of the two largest underpenetrated addressable markets in the group, second only to Indonesia in population, a growing base of confirmed Chinese commercial engagement, and demand side fundamentals that are arguably better documented than either alternative, specifically the running cost arithmetic in section 4 and the storage import trajectory in section 5. What it does not offer is comparable policy certainty, and it carries two burdens that sections 5.4 and 6.3 examine: average industrial power prices well above those in India and China, only partly offset for new plants by a three year incremental tariff, and the most recent record of import and repatriation controls in the group. The battery manufacturing policy remains under discussion in the cabinet. The battery specific instrument remains unfinished. The successor auto policy has only the Prime Minister's approval in principle. An investor placing high value on regulatory predictability, with flexibility to deploy elsewhere, would reasonably weight that heavily and might conclude the uncertainty is inadequately compensated.

This report does not contend that Pakistan should simply be discounted. An investor prioritising early mover positioning and prepared to accept policy risk in exchange for a foothold in the least contested of the five markets has sound grounds for treating the gap between ambition and notified law as the opportunity rather than solely as the risk. No Chinese manufacturer has yet reached volume production of battery electric vehicles in Pakistan; plug-in hybrid assembly has begun, and BYD's Gharao plant has yet to start. CATL remains at discussion stage. The competitive field is materially thinner than in any of the four comparators, and it will not remain so once the policy framework settles.

The structure that reconciles these positions is the one recommended throughout this report: commitments should be made against notified rather than proposed policy; measurable technical milestones should be required; renegotiation triggers should be built into terms signed ahead of official policy notification and enforcement; tariff based incentives should be preferred over disbursement based ones given the delivery record; and the licensing precedent should be followed rather than the equity precedent where market access rather than deep capability is the objective. None of these measures eliminates Pakistani policy risk, and nothing available to a foreign investor could eliminate it entirely. What they do is convert an unhedged bet on a pitchbook into a staged

position where each increment of capital is matched to an increment of legal certainty that has actually arrived, which is the most any investor can reasonably ask of a market at this stage of development.

8 Findings

Seven findings follow from the analysis. Each is stated with the evidentiary standard on which it rests, in the terms defined in section 2.2.

8.1 The framework governing battery manufacturing is unfinished

Of the six instruments governing the sector, three are notified and enforceable, two rest on agreement in principle, and one exists only in draft. None of the three enforceable instruments governs battery manufacturing, and all four incentive lines proposed for battery manufacturing exist only in an investment pitchbook. Every measure that would determine the tariff treatment, local content obligation or incentive entitlement of a battery project in Pakistan remains in principle, in draft or only in promotional material. A financial model built on the 70% and 80% localisation targets or the 0% cell import duty is underwriting an instrument that does not yet exist. This is a documented fact about the legal status of the instruments, not a projection.

8.2 The only track record shows 13% delivery against target

The PAVE pilot targeted 41,000 vehicles and delivered 5,409 by May 2026, 13% of its target and about 4.5% of the full year target of 119,170, despite a Rs. 9 billion budget allocation, political priority and administration by an established body. The diagnosed cause was neither demand nor manufacturing capacity but a payment design that placed the financing burden on the buyers least able to carry that burden: banks approved 4,075 of 44,689 lease applications. The correction, reversing the payment flow, was approved within roughly nine months. Both halves matter: the shortfall establishes an empirical discount factor for headline targets in this sector, and the correction establishes that the review machinery functions rather than merely existing on paper.

8.3 Current Pakistani activity is assembly, not manufacturing

Applying the classification test set out in section 5.1, Pakistan has documented activity at completely built unit import, completely knocked down assembly and, emerging, pack assembly. It has no documented cell production anywhere in the country. The first dedicated lithium battery facility, at Korangi, runs at roughly 4 MW initial capacity and about 2,000 units a month for small vehicle batteries, which is a genuine first mover and a proof of concept but not a scaled

industry. The same slippage between description and substance runs through the vehicle side, where facilities publicly described as manufacturing plants are performing assembly functions.

8.4 The mineral case is sound in geology, unproven in metallurgy

The rationale for prioritising lithium iron phosphate rests on Pakistan's domestic iron and phosphate base. That claim is substantially true at the level of geological occurrence for iron and phosphate, weaker for manganese, and essentially untrue for lithium and graphite in any near term commercial sense. No public investment material or survey document reviewed for this report establishes that battery-grade processing of Pakistani iron or phosphate has been proven, let alone scaled. The existing phosphate base was built and optimised for fertiliser grade output over decades, and converting or paralleling it to battery grade is a substantial capital and engineering undertaking that has not begun at meaningful scale.

8.5 Chinese engagement is real but concentrated at assembly

Six named engagements could be identified. Three are confirmed operations: the Gharo CKD plant, financially closed at approximately USD 150 million for 25,000 units a year, with BYD supplying technology and branding under licence rather than taking equity, so that the balance sheet exposure sits with the two Pakistani partners; and two Chery-linked plug-in hybrid assembly operations, one of them at a USD 100 million plant in Faisalabad. The battery side consists of one memorandum with a CATL-backed company and one government-level conversation with CATL itself, with no disclosed figure, capacity or timetable. Describing the memorandum as CATL entering Pakistan would misstate the counterparty.

8.6 Headline statistics do not survive their own denominators

The widely circulated 61% monthly increase in electric vehicle sales describes a movement from 33 to 53 units, against roughly 15,500 conventional vehicles sold in the same month, in a month when conventional sales were themselves unusually weak. The following month the most visible model fell 43%. The published dataset is too fragmented to support any claim about trajectory beyond the existence of a real demand signal. By contrast, the running cost advantage does survive examination: at current petrol prices the saving at public chargers is roughly 38% to 83%, depending on the operator and charger type, and the more dramatic figure in circulation describes home solar charging. The national scale figures in the policy announcement do not survive as reported: the 126 TWh attributed to electric vehicles is the policy document's figure for national electricity use, against projected electric vehicle demand of about 1.5 TWh, and the Rs.

174 billion to Rs. 105 billion reduction concerns capacity payments, not the fuel import bill.

8.7 Pakistan offers large upside and the least certainty

Indonesia leads on existing ecosystem and committed Chinese capital, Vietnam on policy certainty and charging, and Bangladesh on net incentive value for large groups. The global minimum tax neutralises much of the income tax advantage offered by Vietnam, Indonesia and Thailand to groups of CATL's and BYD's scale. Pakistan matches Indonesia on market upside and has the thinnest competitive field, but scores lowest on policy certainty and near the bottom on ecosystem and charging, and it carries average industrial power prices well above those of India and China and a recent record of import and repatriation controls. The comparison resolves into a trade-off rather than a ranking.

9 Discussion and Implications

The findings above establish a position at a single point in time. Their practical value depends on what would change them, and on what an investor can defensibly do while they hold.

9.1 The Triggers That Will Resolve the Open Questions

Because so much of this report turns on the distinction between notified and proposed, the most useful practical contribution it can make beyond analysis at a single point in time is specifying exactly what would resolve each open question. Several of these open questions have defined triggers rather than remaining permanently uncertain.

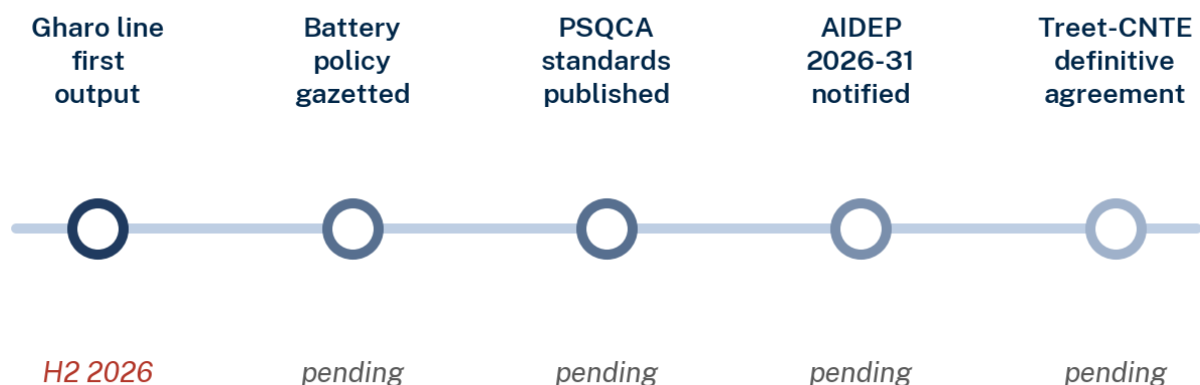


Figure 16. Four of the five decisive events remain pending; the first is scheduled and will arrive before the rest

The five triggers that would resolve the report's principal open questions, with status at September 2026
 Source: ARS analysis. Gharo timing from Khan.⁶³

Notes: The Gharo timetable has already slipped once, from the first half of financial year 2026 to the second half of calendar 2026.

Gazetting of the National Lithium-Ion Battery Manufacturing Policy, converting the 70-80% targets and the 0% cell duty from pitchbook figures into enforceable law, is the single most consequential trigger for the battery investment case. Its timing determines when a staged approach can move from its initial notified tranche into the cell and material processing stages that depend on the finalised schedule. Publication of the finalised battery testing and certification standards is closely related, since these determine the capital expenditure and qualification timeline any manufacturer must budget for, and their absence means even a manufacturer ready to invest cannot yet plan against a known compliance target.

On the vehicle side, formal notification of the successor auto policy, converting the 40%, 45% and 90% value addition thresholds approved in principle in September 2026 into gazetted law, is the equivalent trigger. The percentages that survive that process, and whether the reduction of the weighted average import tariff to 5.99% by 2030 survives intact, will materially affect the economics of every assembly venture currently operating.

Beyond these, the most useful real time indicator of whether localisation is actually occurring rather than merely being promised would be Pakistani customs

import data broken down by battery component category, distinguishing among cells, modules, finished packs, cathode materials and anode materials. This granularity does not currently exist in published trade statistics. Establishing such granularity, whether through direct engagement with the Federal Board of Revenue or through modelling from shipment level trade data, would do more than any other single effort to separate genuine progress from assembly plants presented as manufacturing units.

Two further triggers provide the clearest concrete test of the distinction running through this report. The first is the actual local content composition of vehicles from the Gharo line once commercial production begins, which offers the first real world test of whether Pakistan's most advanced Chinese linked project is a step toward manufacturing or remains assembly in substance regardless of description. That test has now been deferred at least once, and the revised timetable should itself be treated as evidence bearing on the execution risk assessed in section 3.2. The second is whether the Treet relationship progresses beyond memorandum into a definitive agreement covering actual local assembly or value addition, which would be the first concrete test of whether CATL linked technology is moving toward industrialisation or remaining exploratory. Secondary city and corridor charging deployment outside the three major cities is worth tracking as an independent indicator of whether the market is broadening geographically or remaining concentrated in a narrow urban band.

9.2 What Follows for Policy

Four implications follow for Pakistani policymakers, and they are stated here because the investment case and the policy design are not separable questions.

The first concerns sequencing. The tariff structure intended to protect future domestic manufacturing is, in the interim, sustaining elevated pack prices and retarding the adoption curve the wider policy is trying to accelerate. That tension admits of no costless resolution, but it is currently being managed implicitly rather than addressed. A published transition schedule, stating when protection rises and against what observed localisation, would convert an implicit trade-off into a legible, published commitment.

The second concerns the sunset mechanism identified in Bangladesh's design. Pakistan's framework contains no equivalent bounded window on component concessions. Without one, the escalating tariff perimeter described in section 3.4 depends entirely on the administrative discretion that updates the localised parts schedule, which is a weaker commitment device than a date fixed in the instrument itself.

The third concerns the delivery chain rather than the policy text. The PAVE record establishes that Pakistan's binding constraint in this sector is not policy ambition but administrative execution. Every incentive designed as a disbursement inherits that constraint; every incentive designed as a tariff concession does not. A framework that shifted its weight toward border-operated instruments would deliver more of what it promises, at the same headline cost, without requiring any improvement in administrative capacity.

The fourth concerns power. Cell and material processing are among the most power-intensive stages of the chain, and the incremental industrial tariff introduced in October 2025 runs for three years. A power price commitment for battery manufacturing that outlasts a cell line's payback period would do more for the cell ambition than any figure in the pitchbook.

10 Conclusion

Considering in turn the policy architecture, the market data, the supply chain, the mineral endowment, the named tracker and the regional comparison, the conclusion this report reaches is that the demand conditions for a Pakistani battery and electric vehicle industry are real, and that on the evidence assembled they are stronger than the level of international attention this market attracts. The increase in storage imports of over 200% year on year in 2025, a running cost advantage of roughly 38% to 83% against petrol at reported public charging prices, the country's dependence on fuel imports, its notified vehicle policy funded by a dedicated levy rather than an annual budget fight, and credible named Chinese engagement spanning assembly, battery memoranda and government to government discussion all point in the same direction.

What is not there is the industrial depth to convert that demand into a domestic manufacturing base, and it will not arrive quickly regardless of how the pending policies are finalised, given the multi year timelines that cell manufacturing, battery-grade mineral processing and genuine technology transfer have required in every comparable market examined here. Pakistan has no cell manufacturing. Its mineral endowment is geologically real but metallurgically unproven. Its charging network is two orders of magnitude below its own stated target. Its most recent subsidy programme delivered 13% of its pilot target before being redesigned, despite its budget allocation. Its average industrial power price is well above India's and China's, and within the past four years it has restricted imports and profit repatriation.

The most investable opportunities available to a Chinese partner today are consequently not the cell gigafactory propositions that occasionally circulate in Pakistani coverage. They are the more modest and more immediately achievable layers: pack and module assembly, battery management system integration, thermal systems, testing and certification infrastructure, recycling, charging network development, and battery-grade mineral processing specifically targeted at the iron and phosphate base that is geologically present though commercially unproven. Each of these has a real addressable market today, each is consistent with demonstrated Pakistani capability, and each positions an investor to move up the ladder as the policy framework settles rather than betting on a rung that does not yet exist.

Cell manufacturing remains the strategic prize at which both Pakistani policymakers and the Chinese investors evaluating this market are, reasonably, aiming. Reaching it will require policy certainty Pakistan has not yet delivered, quality standards it has not yet finalised, a reliable power supply, committed technical partners beyond the memoranda currently in place, and upstream processing capability it has not yet built. That is a long list. It is not, however, an impossible one, and the direction of travel across every section of this report points toward those conditions. The question confronting an investor is not whether Pakistan will eventually arrive at that position. It is whether the option value of early placement, in the least contested of five competing markets, adequately compensates for underwriting a policy framework still in the course of being written. Reasonable investors will reach different answers, and this report has deliberately declined to assert a single correct answer, since the available evidence does not settle the question.

Appendix A. Evidentiary Classification of the Instruments and Incentive Lines

This appendix sets out the classification underlying Figure 1 and Figure 3 and Finding 8.1, instrument by instrument, so that each allocation can be tested against its stated evidence rather than accepted on the aggregate.

Table 1. The six governing instruments, by legal standing at September 2026

Instrument	Legal standing	Basis for the classification
New Energy Vehicle Policy 2025-30	Notified	Approved by the Economic Co-ordination Committee in August 2025 and in force
New Energy Vehicle Adoption Levy 2025	Notified	Levy of 1% to 3% on new petrol and diesel vehicles under the Finance Act 2025, financing the policy's incentives
CKD tariff tiers for EV assembly, at 1%, 10% and 25%	Notified	Inherited from the expired auto policy and extended in the 2026-27 budget to June 2027
National Lithium-Ion Battery Manufacturing Policy	In principle	Working group constituted 12 December 2025, chemistry approved, forwarded to the Ministry and awaiting the National Tariff Board, the Prime Minister and cabinet
Successor auto policy, AIDP 2026-31	In principle	Prime Minister's approval in principle on 9 September 2026; awaiting the Economic Coordination Committee, cabinet and IMF review
EV Battery Policy	Draft	Publicly committed to by officials; no published text, and the standards and storage policy it depends on still in development

Source: ARS analysis of the sources cited in section 3.

Notes: The incentive lines proposed for battery manufacturing are classified separately in Table 2, because they are proposed components of an unfinished instrument rather than instruments in their own right.

Table 2. The four incentive lines proposed for battery manufacturing, by legal standing at September 2026

Incentive line	Proposed terms	Legal standing
Customs duty on imported cells	0% to 2028, rising gradually to about 10% thereafter	Pitchbook only
Federal excise duty on imported completely built batteries	20%	Pitchbook only
Sales tax for local battery manufacturers	1%	Pitchbook only
Accelerated depreciation	Up to 90% in the first year	Pitchbook only

Source: ARS analysis of Engineering Development Board.¹⁶

Appendix B. Named Chinese Engagements by What They Commit

This appendix sets out the tracker of six engagements underlying Figure 14 and Finding 8.5. Engagements are ordered by commitment level, not by the prominence of the announcement.

Table 3. Named Chinese engagements in Pakistan’s EV and battery sector, September 2026

Counterparties	Activity	Commitment level
Hub Power Holdings and Mega Conglomerate, through Mega Motor Company, with BYD under licence	CKD vehicle assembly plant at Gharo, Sindh, approximately USD 150 million for 25,000 units a year	Confirmed and funded, financially closed January 2026; launch delayed to the second half of 2026
Chery and Master Group	Locally assembled Tiggo 8 plug-in hybrid launched January 2026; Tiggo 9 plug-in hybrid on sale	Confirmed and producing
Jaecoo and Omoda, through Nishat’s NexGen Auto	Local production of the Jaecoo J7 plug-in hybrid at Faisalabad from October 2025, at a USD 100 million plant	Confirmed, funded and producing
Treet Battery Limited and Contemporary Nebula Technology Energy, in which CATL holds a strategic investment	Framework to explore energy storage cooperation and introduction of CATL cell-based products	Signed memorandum, 25 May 2026, scope undefined
StarCharge Energy Pakistan and Bahum Global	Charging ecosystem development with stated local manufacturing and technology transfer ambitions	Signed memorandum, July 2026, scope and timeline undefined
Government of Pakistan and CATL	Discussions on battery technology, storage and investment across renewable energy and electric mobility	Government-level talks only, no disclosed figure, capacity or timetable

Source: ARS analysis of the sources cited in section 6.

BYD holds no equity in Mega Motor Company. Its participation runs through a master supply and manufacture agreement, a technical licence and a distribution arrangement. Jetour’s assembly through United Motors is of petrol models and is not counted.

Appendix C. Scoring Rubric for the Analytical Figures

Figures 6, 13 and 15 present analytical scores. They are ordinal judgments on the evidence reviewed, not measurements. This appendix sets out the scale anchors used for each figure and the evidence behind every score, so that a reader who weighs the evidence differently can re-score it and see how the conclusion would change.

Figure 6 scores Pakistani capability in each layer of the vehicle and battery value chain on a scale of 0 to 100, anchored as follows: 0 means no documented activity; 10 to 20, pilot or single firm activity dependent on imported design and core

components; 30 to 40, several firms producing with imported core components; 50 to 60, established domestic supply for most parts with imported subcomponents; 80 to 100, established domestic production with local design capability and supplier depth. The layers scored 0 sit on the same scale as the others: no documented activity is the bottom anchor, not a separate category.

Table 4. Evidence behind the capability scores in Figure 6

Value chain layer	Score	Evidence
Body and chassis	85	Decades of local body, chassis and sheet metal production for cars; over 90% local content in two and three wheelers (section 3.1)
Wiring and thermal	55	Established wiring harness and cooling system supply for conventional vehicles; electric thermal management not documented
Module and pack	35	Korangi facility and pack assembly of imported cells by several firms (section 5.1)
Motor and inverter	15	No documented production of automotive traction motors or inverters; the score reflects limited local assembly of motors and controllers for electric two wheelers
Battery management system	10	Integration of imported systems in pack assembly; no documented local design
Cells	0	No documented cell production (section 5.1)
Cathode and anode materials	0	No documented production (section 5.3)

Figure 13 scores each mineral input on two scales of 0 to 100. For geological evidence: 0 to 20, exploration reports only; 20 to 40, documented occurrences with limited survey coverage; 40 to 60, multiple documented occurrences with inconsistent reserve estimates; 60 to 80, established mining; 80 to 100, established mining at several sites with published survey coverage. For battery-grade processing readiness: 0 to 10, no processing beyond extraction; 10 to 20, limited processing for other uses; 20 to 40, established processing for another industry with no documented route to battery grade; 40 to 70, pilot battery-grade processing; 70 to 100, commercial battery-grade output.

Table 5. Evidence behind the mineral scores in Figure 13

Input	Geology	Processing	Evidence
Iron	88	30	Deposits at Kalabagh, Mianwali, Chiniot and in Balochistan; processing to steelmaking grade only
Phosphate	80	25	Established mining in Hazara and Abbottabad; processing to fertiliser grade only
Manganese	55	12	Scattered occurrences with inconsistent reserve estimates; limited processing
Graphite	40	8	Occurrences in northern Pakistan and Khyber Pakhtunkhwa with quality and scale constraints

Input	Geology	Processing	Evidence
Lithium	22	3	Exploration stage in Gilgit-Baltistan and Azad Jammu and Kashmir

Figure 15 scores the five countries from 0 to 10 on six dimensions, using the anchors in Table 6. Scores are not summed, because the dimensions do not carry equal weight for every investor. Table 7 gives the score and its basis for each country. Where national data are thin, as for charging networks outside Pakistan, the score is a banded judgment and the table says so.

Table 6. Scale anchors for the comparative scorecard in Figure 15

Dimension	Scale anchors
Market size and upside	9 to 10: population above 200 million and electric share of new sales below 5%. 7 to 8: population of 150 to 200 million with share below 5%. 5 to 6: population of 50 to 150 million with share below 20%. Lower scores for smaller or more saturated markets.
Existing ecosystem	9 to 10: volume vehicle assembly by several makers and operating cell production. 7 to 8: volume assembly and a cell plant operating or under construction. 5 to 6: volume assembly with pack assembly. 3 to 4: small scale assembly and pack assembly of imported cells. 1 to 2: import led.
Policy certainty	9 to 10: incentives in statute with dated duration and no recent reversal. 7 to 8: incentives in regulation or budget law with dated duration. 5 to 6: codified but with material recent expiries or reversals. 3 to 4: a mix of notified and pending instruments. 1 to 2: core instruments pending or in draft.
Chinese capital committed	9 to 10: over USD 1 billion, including cell or materials projects. 7 to 8: USD 250 million to USD 1 billion, or any funded cell project. 5 to 6: USD 100 to 500 million in funded vehicle assembly. 3 to 4: memoranda and projects under USD 100 million. 1 to 2: none documented.
Charging infrastructure	7 to 10: thousands of public points and a funded national network under way. 5 to 6: a national network under construction against a dated target. 3 to 4: hundreds of points. 1 to 2: fewer than 100 points or none documented.
Net incentive value	Value to a group with revenue above EUR 750 million after the global minimum tax. 8 to 10: large duty, excise or cash incentives in force and unaffected. 6 to 7: substantial non-income incentives in force, income tax holidays neutralised. 4 to 5: incentives mainly income tax holidays, or narrow notified incentives with the main ones pending. 1 to 3: little in force.

Table 7. Scores and evidence for the comparative scorecard in Figure 15

Dimension	Pakistan	Bangladesh	Vietnam	Indonesia	Thailand
Market size and upside	9: about 240 million people; electric share of car sales under 1% (section 4.1)	7: about 170 million; 669 registered electric vehicles	6: about 100 million people	9: about 280 million; electric share 2.9% in mid 2024 ⁸⁹	5: about 70 million; electric share 17.7% in 2025
Existing ecosystem	3: plug-in hybrid assembly; pack assembly of imported cells	2: import led; no vehicle or cell plant documented	8: domestic volume maker; 5 GWh Gotion and VinES cell plant ⁹⁸	9: 10 GWh cell plant operating; BYD and other vehicle plants	8: BYD and other plants; first cell plant approved 2025

Dimension	Pakistan	Bangladesh	Vietnam	Indonesia	Thailand
Policy certainty	2: battery instruments unfinished; hybrid tax changed twice in 2026	7: budget instrument with dated sunset	9: incentives in a new statute from 2025	5: import incentive ended 2025; government ended the LG project	7: EV 3.5 codified to 2027; successor regime unsettled
Chinese capital committed	5: USD 150 million Gharo and USD 100 million Faisalabad, Chinese technology with Pakistani capital	1: none documented	7: Gotion cell plant of about USD 275 million	10: CATL USD 6 billion chain; BYD plant	9: BYD USD 490 million; Sunwoda over USD 1 billion in cells
Charging infrastructure	2: over 20 public points	1: none documented	7: banded; share of about 24,000 chargers across four South-east Asian markets ⁹⁹	5: banded; target of 30,000 by 2030	6: banded; target of 12,000 by 2030
Net incentive value	4: 1% sales tax and CKD tiers notified; battery incentives pending	8: duty tiers by value addition; no minimum tax regime documented	5: income tax holidays neutralised by Decree 236	6: value added tax and import concessions; holidays neutralised by PMK 136	7: cash subsidy and 2% excise; holidays limited by top-up tax

Source: ARS analysis of the sources cited in sections 3 to 7 and in this appendix.

Glossary

AIDEP

Auto Industry Development and Export Policy. The 2021-26 instrument expired on 30 June 2026; its successor, the Automotive Industry Development Policy (AIDP) 2026-31, received the Prime Minister's approval in principle on 9 September 2026 but had not been notified at the date of this report.

BDT

Bangladeshi taka.

BMS

Battery management system. The electronics and software controlling cell balancing, thermal management and state-of-charge estimation in a battery pack.

CATL

Contemporary Amperex Technology Company Limited, the world's largest battery manufacturer by volume.

CBU

Completely built unit. A vehicle imported fully assembled.

CKD

Completely knocked down. A vehicle imported as a kit of parts for local assembly.

CNTE

Contemporary Nebula Technology Energy Company, a separate company in which CATL holds a strategic investment.

ECC

Economic Coordination Committee of the Cabinet, Government of Pakistan.

EFF

Extended Fund Facility, the International Monetary Fund lending arrangement approved for Pakistan in September 2024.

EDB

Engineering Development Board, the body administering the PAVE subsidy scheme.

FED

Federal excise duty.

GMT

Global minimum tax. The 15% minimum effective tax rate applied to multinational groups with consolidated revenue of at least EUR 750 million.

GWh

Gigawatt hour, a unit of energy. Not interchangeable with GW, a unit of capacity.

IEEFA

Institute for Energy Economics and Financial Analysis.

kWh

Kilowatt hour, a unit of energy.

LFP

Lithium iron phosphate, the battery chemistry approved for first-phase localisation in Pakistan.

MW

Megawatt, a unit of capacity.

NEPRA

National Electric Power Regulatory Authority, Pakistan's electricity regulator.

NEECA

National Energy Efficiency and Conservation Authority, which licenses electric vehicle charging infrastructure.

NEV

New energy vehicle. In the Pakistani policy framework the term covers battery electric and plug-in hybrid vehicles.

PACRA

Pakistan Credit Rating Agency.

PAMA

Pakistan Automotive Manufacturers Association.

PAVE

Pakistan Accelerated Vehicle Electrification, the subsidy programme under the New Energy Vehicle Policy 2025-30, financed through the New Energy Vehicle Adoption Levy.

PHEV

Plug-in hybrid electric vehicle.

PSQCA

Pakistan Standards and Quality Control Authority, the body developing the battery testing, certification and recycling regime.

SRO

Statutory regulatory order, the instrument through which the schedule of localised parts is periodically revised.

TWh

Terawatt hour, a unit of energy.

References

- 1 Renewables First. 2026. *From Solar Panels to Storage: Pakistan's Battery Boom Begins*. Renewables First, July 2026. <https://www.renewablesfirst.org/>.
- 2 Ember. 2026. "The Solarisation of Pakistan's Energy Economy." Ember, 25 June 2026. <https://ember-energy.org/latest-insights/the-solarisation-of-pakistans-energy-economy/>.
- 3 Vistra. 2026. "India Eases FDI Rules for Countries Sharing Land Borders with India." Vistra, 17 March 2026. <https://www.vistra.com/insights/cabinet-decision-changes-fdi-guidelines-countries-sharing-land-borders-india-press-note>.
- 4 Press Information Department. 2025. "New Energy Vehicle (NEV) Policy 2025-30." Government of Pakistan, 19 June 2025. https://pid.gov.pk/site/press_detail/29435.
- 5 The Nation. 2025. "ECC Approves Summary Pertaining to New Energy Vehicle Policy 2025-30." The Nation, 20 August 2025. <https://www.nation.com.pk/20-Aug-2025/ecc-approves-summary-pertaining-to-new-energy-vehicle-policy-2025-30>.
- 6 Ministry of Industries and Production. 2025. *New Energy Vehicles Policy 2025-30*. Government of Pakistan, 13 August 2025. <https://moip.gov.pk/SiteImage/Policy/NEV%20Policy%20Final%2013.8.25.pdf>.
- 7 PakWheels. 2025. "Pakistan Goes Electric: What the New Energy Vehicle Policy 2025-30 Means." PakWheels Blog, 27 August 2025. <https://www.pakwheels.com/blog/pakistan-goes-electric-what-the-new-energy-vehicle-policy-2025-30-means/>.
- 8 Federal Board of Revenue. 2025. Rate of New Energy Vehicle Adoption Levy, First Schedule to the Finance Act 2025. FBR, July 2025. <https://download1.fbr.gov.pk/Docs/2025717157305886EnergyVehicleLevy.pdf>.
- 9 Profit by Pakistan Today. 2026. "Pakistan Accelerated Vehicle Electrification Scheme Launched to Boost Subsidised EV Adoption." Profit, 22 January 2026. <https://profit.pakistantoday.com.pk/2026/01/22/pakistan-accelerated-vehicle-electrification-scheme-launched-to-boost-subsidised-ev-adoption/>.

- 10 Profit by Pakistan Today. 2026. "After Weak Bank Response to Pilot Scheme, Govt Launches Second Phase of PAVE Programme for Subsidised E-Bikes, Rickshaws." Profit, 6 May 2026. <https://profit.pakistantoday.com.pk/2026/05/06/after-weak-bank-response-to-pilot-scheme-govt-launches-second-phase-of-pave-programme-for-subsidised-e-bikes-rickshaws>.
- 11 Profit by Pakistan Today. 2026. "Pakistan Advances National Lithium-Ion Battery Manufacturing Policy." Profit, 7 January 2026. <https://profit.pakistantoday.com.pk/2026/01/07/pakistan-advances-national-lithium-ion-battery-manufacturing-policy/>.
- 12 Geo News. 2026. "PM Approves Auto Policy 2026-31 Linking Incentives to Exports, Localisation." Geo News, 10 September 2026. <https://www.geo.tv/latest/681475-pm-approves-auto-policy-2026-31-linking-incentives-to-exports-localisation>.
- 13 ProPakistani. 2026. "Govt Approves 2nd Phase of Vehicle Electrification Plan for Cheaper Bikes, Rickshaws." ProPakistani, 5 May 2026. <https://propakistani.pk/2026/05/05/govt-approves-new-subsidy-plan-under-ev-policy/>.
- 14 ProPakistani. 2026. "Banks Approve Only 4,075 of 44,689 E-Bike, Rickshaw Loan Applications Under PM Scheme." ProPakistani, 6 May 2026. <https://propakistani.pk/2026/05/06/banks-approve-only-4075-of-44689-e-bike-rickshaw-loan-applications-under-pm-scheme/>.
- 15 Arab News. 2026. "Pakistan Moves to Localize Lithium-Ion Battery Manufacturing under New Energy Security Policy." Arab News Pakistan, 7 January 2026. <https://www.arabnews.pk/node/2628539/pakistan>.
- 16 Engineering Development Board. 2026. *China-Pakistan Investment Pitch Book: Battery Energy Storage Systems, Power Storage and Advanced Batteries*. Engineering Development Board, Government of Pakistan, 2026. Published by the Consulate General of Pakistan, Guangzhou. https://www.pakcggz.gov.pk/trade-and-investments/investment-pitchbook-file/1778555940_6a029c24ac8f6.pdf.
- 17 Arab News. 2026. "Pakistan Woos Chinese Investors to Manufacture Lithium-Ion Battery Cells Locally." Arab News Pakistan, 2 July 2026. <https://www.arabnews.pk/node/2649357/pakistan>.
- 18 Dawn. 2026. "Ministry Pushes EV Battery Policy." Dawn, 18 August 2026. <https://www.dawn.com/news/2023581/ministry-pushes-ev-battery-policy>.
- 19 Engineering Development Board. 2021. *Auto Industry Development and Export Policy 2021-26*. Ministry of Industries and Production and Engineering Development Board, Government of Pakistan, 2021. <https://moip.gov.pk/SiteImage/Policy/Auto%20Industry%20Development%20and%20Export%20Policy%202021-26.pdf>.
- 20 Dawn. 2026. "Duty Raised on EVs Priced above Rs20m in FY27." Dawn, 13 June 2026. <https://www.dawn.com/news/2007451/duty-raised-on-evs-priced-above-rs20m-in-fy27>.
- 21 Business Recorder. 2026. "Govt Cuts HEV Sales Tax from 25% to 18%." Business Recorder, 14 September 2026. <https://www.brecorder.com/news/40439415/govt-cuts-hev-sales-tax-from-25-to-18>.

- 22** Profit by Pakistan Today. 2026. "Hybrid Vehicles Face 25% Sales Tax from July 1." Profit, 7 July 2026. <https://profit.pakistantoday.com.pk/2026/07/07/hybrid-vehicles-are-now-subject-to-25percent-sales-tax>.
- 23** Arab News. 2026. "Pakistan Says New Auto Policy Nears Approval as Tariff Cuts Divide Industry." Arab News Pakistan, 6 August 2026. <https://www.arabnews.pk/node/2653585/pakistan>.
- 24** ProPakistani. 2026. "Ishaq Dar Led Committee Gives In-Principle Approval to New Auto Policy." ProPakistani, 13 August 2026. <https://propakistani.pk/2026/08/13/ishaq-dar-led-committee-gives-in-principle-approval-to-new-auto-policy/>.
- 25** Profit by Pakistan Today. 2026. "Pakistan's Auto Policy Nears Final Approval, but Tariff Cuts Remain Biggest Sticking Point." Profit, 7 August 2026. <https://profit.pakistantoday.com.pk/2026/08/07/pakistans-auto-policy-nears-final-approval-but-tariff-cuts-remain-biggest-sticking-point>.
- 26** ProPakistani. 2026. "PM Gives In-Principle Approval to New Auto Policy 2026-31." ProPakistani, 9 September 2026. <https://propakistani.pk/2026/09/09/pm-gives-in-principle-approval-to-new-auto-policy-2026-31/>.
- 27** ProPakistani. 2026. "Petrol Car Sales Decrease by 9% and Electric Car Sales Increase by 61% in Just 1 Month in Pakistan." ProPakistani, 13 April 2026. <https://propakistani.pk/2026/04/13/petrol-car-sales-decrease-by-9-and-electric-car-sales-increase-by-61-in-just-1-month-in-pakistan/>.
- 28** PakWheels. 2026. "Pakistan Car Sales April 2026: Model-Wise Sales Breakdown." PakWheels Blog, 13 May 2026. <https://www.pakwheels.com/blog/pakistan-car-sales-april-2026-model-wise-sales-breakdown/>.
- 29** PakWheels. 2026. "New Car Prices in Pakistan." Ex-factory price listings for BYD, GWM Haval, Toyota and Suzuki models, accessed 24 September 2026. <https://www.pakwheels.com/new-cars/>.
- 30** Daily Pakistan. 2026. "New Petrol, Diesel Prices in Pakistan for September 24, 2026." Daily Pakistan, 23 September 2026. <https://en.dailypakistan.com.pk/23-Sep-2026/new-petrol-diesel-prices-in-pakistan-for-september-24-2026>.
- 31** National Electric Power Regulatory Authority. 2025. Decision of the Authority regarding Rationalization of Tariff for Electric Vehicle Charging Stations. NEPRA, 15 April 2025. <https://www.nepra.org.pk/tariff/Tariff/Ex-WAPDA%20DISCOS/2025/TRF-100%20EV%20RATIONALIZATION%20OF%20TARIFF%20FOR%20EV%20CHARGING%20STATION%2015-04-2025%205469-72.pdf>.
- 32** Profit by Pakistan Today. 2025. "NEPRA Slashes EV Charging Station Tariff by Rs22." Profit, 16 April 2025. <https://profit.pakistantoday.com.pk/2025/04/16/nepra-slashes-ev-charging-station-tariff-by-rs22>.
- 33** The News. 2025. "Nepra Cuts EV Charging Tariff by 45%, Backs Cross-Subsidy Plan." The News, 16 April 2025. <https://www.thenews.com.pk/print/1301809-nepra-cuts-ev-charging-tariff-by-45pc-backs-cross-subsidy-plan>.

- 34 PakWheels. 2026. "Lahore EV Charging Guide 2026: Costs, Stations, and Savings Explained." PakWheels Blog, 3 April 2026. <https://www.pakwheels.com/blog/lahore-ev-charging-guide-2026-costs-stations-savings/>.
- 35 ChargePK. 2026. "How Much Does It Cost to Charge an EV in Pakistan? (2026)." ChargePK Blog, 6 July 2026. <https://chargepk.com/blog/how-much-does-it-cost-to-charge-an-ev-in-pakistan.html>.
- 36 Renewables First. 2025. *Pakistan Electricity Review 2025*. Renewables First, 2025. https://uploads.renewablesfirst.org/Pakistan_Electricity_Review_2025_89f0b613d6.pdf.
- 37 Arab News. 2026. "Pakistan Plans 3,000 EV Charging Stations as Green Mobility Push Gathers Pace." Arab News Pakistan, 14 January 2026. <https://www.arabnews.pk/node/2629341/pakistan>.
- 38 Business Recorder. 2025. "From Karachi to Peshawar: HUBCO Green Accelerates Pakistan's EV Charging Rollout." Business Recorder, 30 October 2025. <https://www.brecorder.com/news/40390045>.
- 39 Associated Press of Pakistan. 2025. "PSO, BYD, HUBCO Green Expand NEV Charging Network along Lahore-Islamabad Motorway." APP, 25 June 2025. <https://www.app.com.pk/national/pso-byd-hubco-green-expand-nev-charging-network-along-lahore-islamabad-motorway-2/>.
- 40 Dawn. 2026. "First Lithium Battery Plant to Be Set Up in Korangi." Dawn, 12 April 2026. <https://www.dawn.com/news/1990901>.
- 41 Profit by Pakistan Today. 2026. "Pakistan's First Lithium-Ion Battery Plant Set to Begin Production as New Industrial Policy Nears Approval." Profit, 12 April 2026. <https://profit.pakistantoday.com.pk/2026/04/12/pakistans-first-lithium-ion-battery-plant-set-to-begin-production-as-new-industrial-policy-nears-approval>.
- 42 Institute for Energy Economics and Financial Analysis (IEEFA). 2025. *Battery Storage and the Future of Pakistan's Electricity Grid*. IEEFA, September 2025. <https://ieefa.org/resources/battery-storage-and-future-pakistans-electricity-grid>.
- 43 pv magazine. 2026. "Pakistan Imports 4.6 GWh of BESS in 2025." pv magazine, 15 July 2026. <https://www.pv-magazine.com/2026/07/15/pakistan-imports-4-6-gwh-of-bess-in-2025/>.
- 44 The Express Tribune. 2026. "Govt Unveils Plan for 90% Clean Electricity." The Express Tribune, 5 August 2026. <https://tribune.com.pk/story/2622122/govt-unveils-plan-for-90-clean-electricity>.
- 45 Mirza, Muhammad Ali. 1995. "An Overview of Mineral Resources of Pakistan." Geological Survey of Pakistan, April 1995. <https://gsp.gov.pk/report/an-overview-of-mineral-resources-of-pakistan-1995/>.
- 46 Abubakar, Muhammad Ashraf, Sajjad Khan and Muhammad Azeem. 2024. "Economic Potential and Health Risks Assessment of the Trace Elements in the Recently Discovered Phosphate Deposits of Paswal Mian, Banseri and Shahkot Regions, Abbottabad, Khyber Pakhtunkhwa, Pakistan." *Geologica* 7 (1). Geological Survey of Pakistan, 2024. <https://geologica.gov.pk/wp-content/uploads/2024/05/GEOLOGICA-v7-Article-2.pdf>.

- 47 Punjab Mineral Company. 2026. “Kalabagh Iron Ore Resource Project.” Punjab Mineral Company, Government of the Punjab, accessed September 2026.
https://pmc.punjab.gov.pk/kalabagh_iron_ore_resource-project.
- 48 DeCarlo, Keita F., and Ji Won Moon. 2025. “The Mineral Industry of Pakistan.” In *2020-2021 Minerals Yearbook*, vol. 3. United States Geological Survey, January 2025.
<https://pubs.usgs.gov/myb/vol3/2020-21/myb3-2020-21-pakistan.pdf>.
- 49 Guberman, David, Samantha Schreiber and Anna Perry. 2024. *Export Restrictions on Minerals and Metals: Indonesia’s Export Ban of Nickel*. Working Paper ICA-104. United States International Trade Commission, February 2024.
https://www.usitc.gov/publications/332/working_papers/ermm_indonesia_export_ban_of_nickel.pdf.
- 50 Degen, Florian, Martin Winter, David Bendig and Jens Tübke. 2023. “Energy Consumption of Current and Future Production of Lithium-Ion and Post Lithium-Ion Battery Cells.” *Nature Energy* 8: 1284-1295. <https://www.nature.com/articles/s41560-023-01355-z>.
- 51 Nukta. 2025. “Pakistan’s Industrial Power Tariff Is among the Highest.” Nukta, 9 March 2025, reporting International Energy Agency, Electricity 2025.
<https://nukta.com/pakistans-industrial-power-tariff-highest>.
- 52 The Express Tribune. 2025. “PM Unveils Power Tariff Cuts for Industry, Farms.” The Express Tribune, 24 October 2025. <https://tribune.com.pk/story/2573961/pm-unveils-power-tariff-cuts-for-industry-farms>.
- 53 Arab News. 2026. “Pakistan Says Power Challenge Has Shifted from Generation to Energy Storage.” Arab News Pakistan, 4 August 2026. <https://www.arabnews.com/node/2653325>.
- 54 Profit by Pakistan Today. 2026. “Discos Bleed Rs265 Billion from T&D Losses, Rs132 Billion from Recovery Shortfalls: NEPRA.” Profit, 16 January 2026.
<https://profit.pakistantoday.com.pk/2026/01/16/nepra-releases-2025-state-of-industry-and-2024-25-annual-reports-for-federal-review>.
- 55 Startup Pakistan. 2021. “More than 50,000 Engineers in Pakistan Are Jobless, Says PEC.” Startup Pakistan, 20 January 2021. <https://startuppakistan.com.pk/more-than-50000-engineers-in-pakistan-are-jobless-says-pec/>.
- 56 ProPakistani. 2025. “57 Companies Get License to Assemble Electric Cars in Pakistan.” ProPakistani, 6 February 2025. <https://propakistani.pk/2025/02/06/57-companies-get-license-to-assemble-electric-cars-in-pakistan/>.
- 57 CnEVPost. 2025. “Thailand Approves Sunwoda’s \$1 Billion Investment in EV Battery Cell Plants.” CnEVPost, 13 March 2025. <https://cnevpost.com/2025/03/13/thailand-approves-sunwoda-investment-ev-battery-cell-plants/>.
- 58 The Jakarta Post. 2024. “Indonesia Launches First EV Battery Plant.” The Jakarta Post, 3 July 2024. <https://www.thejakartapost.com/business/2024/07/03/indonesia-launches-first-ev-battery-plant>.
- 59 Pakistan Credit Rating Agency (PACRA). 2026. *Hub Power Holdings Limited: Rating Report*. PACRA. <https://pacra.com/api/rating-report/MTU3OTk=>.

- 60** Business Recorder. 2026. "HUBCO's Gharo Plant Set to Come Online in Second Half of 2026, Eyes 25,000 Vehicles Annually." Business Recorder, 26 February 2026. <https://www.brecorder.com/news/40409118>.
- 61** Profit by Pakistan Today. 2026. "HUBCO to Launch \$150m Auto Assembly Plant in 2026 as Diversification Drive Accelerates." Profit, 26 February 2026. <https://profit.pakistantoday.com.pk/2026/02/26/hubco-to-launch-150m-auto-assembly-plant-in-2026-as-diversification-drive-accelerates/>.
- 62** HUBCO. 2026. "Mega Motor Company (Private) Limited." The Hub Power Company Limited, accessed September 2026. <https://hubpower.com/mega-motor/>.
- 63** Khan, Aamir Shafaat. 2026. "\$150m BYD Plant Launch in Sindh Delayed Again." Dawn, 6 September 2026. <https://www.dawn.com/news/2027775>.
- 64** Business Recorder. 2026. "CATL-Backed Chinese Firm Eyes Pakistan's Energy Storage Market through Treet Battery Partnership." Business Recorder, 2 June 2026. <https://www.brecorder.com/news/40423615>.
- 65** Associated Press of Pakistan. 2026. "Executive President of CATL, Mr. Oscar Luo Calls on Prime Minister Muhammad Shehbaz Sharif." APP, 24 May 2026. <https://www.app.com.pk/photos-section/executive-president-of-catl-mr-oscar-luo-calls-on-prime-minister-muhammad-shehbaz-sharif/>.
- 66** Business Recorder. 2026. "EV Ecosystem: StarCharge Energy Pakistan, Bahum Global Pakistan Sign MoU." Business Recorder, 21 July 2026. <https://www.brecorder.com/news/40431215>.
- 67** MarkLines. 2025. "Omoda & Jaecoo Begins Local Production of Jaecoo J7 SHS Plug-in Hybrid in Pakistan." MarkLines, 21 October 2025. <https://www.marklines.com/en/news/334564>.
- 68** ProPakistani. 2026. "Chery Tiggo 9 PHEV Price in Pakistan Announced." ProPakistani, 13 February 2026. <https://propakistani.pk/2026/02/13/chery-tiggo-9-phev-price-in-pakistan-announced/>.
- 69** PakWheels. 2025. "New Automotive Players: Companies That Started Local Assembly in Pakistan in 2025." PakWheels Blog, 19 December 2025. <https://www.pakwheels.com/blog/new-automotive-players-local-assembly-pakistan-2025/>.
- 70** Business Recorder. 2023. "Pak Suzuki Extends Plant Shutdown Again due to Inventory Shortage." Business Recorder, 13 January 2023. <https://www.brecorder.com/news/40220057>.
- 71** Business Recorder. 2023. "Foreign Investors: Repatriation of Profit, Dividend Dives to 18-Year Low." Business Recorder, 27 July 2023. <https://www.brecorder.com/news/40254910>.
- 72** The Express Tribune. 2024. "Profit, Dividend Repatriation Rises to Six-Year High." The Express Tribune, 31 July 2024. <https://tribune.com.pk/story/2484324/profit-dividend-repatriation-rises-to-six-year-high>.
- 73** International Monetary Fund (IMF). 2024. "IMF Executive Board Approves 37-Month Extended Arrangement for Pakistan." Press Release 24/343, 27 September 2024.

<https://www.imf.org/en/news/articles/2024/09/27/pr-24343-pakistan-imf-concludes-2024-aiv-consultation-pakistan-approves-37-mo-extended-arr>.

- 74 International Monetary Fund (IMF). 2026. “IMF Executive Board Completes Third Review of the Extended Arrangement under the Extended Fund Facility with Pakistan.” Press Release 26/147, 8 May 2026. <https://www.imf.org/en/news/articles/2026/05/08/pr-26147-pakistan-imf-completes-3rd-rev-of-extended-arrangement-under-eff-and-2nd-rev-arrang-rsf>.
- 75 The News. 2026. “Commitments to IMF Blocking Cut in Electricity Costs: Minister.” The News, 5 August 2026. <https://www.thenews.pk/print/1430258-commitments-to-imf-blocking-cut-in-electricity-costs-minister>.
- 76 The Daily Star. 2026. “Duty Cuts Aim to Boost EV Adoption.” The Daily Star, 11 June 2026. <https://www.thedailystar.net/business/news/duty-cuts-aim-boost-ev-adoption-4196276>.
- 77 Prothom Alo. 2026. “Budget 2026-27: EVs to Get Major Tax Cuts, Petrol and Diesel Cars to Be Costlier.” Prothom Alo English, 11 June 2026. <https://en.prothomalo.com/business/local/tj9vwbv68>.
- 78 Bangladesh Sangbad Sangstha. 2026. “Duty, Tax Exemptions Proposed for Environment-Friendly Battery Industry.” BSS, 11 June 2026. <https://www.bssnews.net/national-budget-2026-2027/395038>.
- 79 The Daily Star. 2026. “EVs Are Trending in Bangladesh. Will It Sustain?” The Daily Star, 26 May 2026. <https://www.thedailystar.net/news/tech-startup/news/evs-are-trending-bangladesh-will-it-sustain-4184956>.
- 80 EY. 2025. “Vietnam Passes New Corporate Income Tax Law.” EY Global Tax Alert, June 2025. https://www.ey.com/en_gl/technical/tax-alerts/vietnam-passes-new-corporate-income-tax-law.
- 81 Vietnam, National Assembly. 2025. *Law on Corporate Income Tax No. 67/2025/QH15*. Socialist Republic of Vietnam, 14 June 2025.
- 82 PwC. 2026. “Vietnam: Corporate Tax Credits and Incentives.” PwC Worldwide Tax Summaries, reviewed 23 September 2026. <https://taxsummaries.pwc.com/vietnam/corporate/tax-credits-and-incentives>.
- 83 Baker McKenzie. 2026. “Vietnam: New Corporate Income Tax Decree.” Baker McKenzie Insight, January 2026. <https://www.bakermckenzie.com/en/insight/publications/2026/01/vietnam-new-corporate-income-tax-decree>.
- 84 VietnamNet. 2026. “Vietnam Extends EV Registration Fee Exemption through 2030.” VietnamNet Global, 10 June 2026. <https://vietnamnet.vn/en/vietnam-extends-ev-registration-fee-exemption-through-2030-2524519.html>.
- 85 EY. 2026. “Vietnam Introduces Super Deduction for R&D Expenses under New Legal Framework.” EY Global Tax Alert, 12 February 2026. https://www.ey.com/en_gl/technical/tax-alerts/vietnam-introduces-super-deduction-for-r-and-d-expenses-under-new-legal-framework.

- 86 Vietnam Briefing. 2025. “Global Minimum Tax Implementation in Vietnam: Decree 236.” Vietnam Briefing, 5 September 2025. <https://www.vietnam-briefing.com/news/global-minimum-tax-implementation-in-vietnam-decree-236.html/>.
- 87 Contemporary Amperex Technology Co. Limited (CATL). 2025. “CATL and Partners Break Ground on US\$6 Billion Battery Integration Project in Indonesia.” CATL, 30 June 2025. <https://www.catl.com/en/news/6481.html>.
- 88 The Diplomat. 2025. “LG’s Exit from a Mega-Battery Project in Indonesia, Explained.” The Diplomat, May 2025. <https://thediplomat.com/2025/05/lgs-exit-from-a-mega-battery-project-in-indonesia-explained/>.
- 89 International Institute for Sustainable Development (IISD). 2025. “Indonesian Electric Vehicle Boom: A Temporary Trend or a Long-Term Vision?” IISD, 7 February 2025. <https://www.iisd.org/articles/deep-dive/indonesian-electric-vehicle-boom-temporary-trend-or-long-term-vision>.
- 90 electrive. 2026. “BYD Inaugurates New Car Factory in Indonesia.” electrive, 8 September 2026. <https://www.electrive.com/2026/09/08/byd-inaugurates-new-car-factory-in-indonesia/>.
- 91 BDO. 2025. “Indonesia: Global Minimum Tax under Pillar Two Implemented.” BDO World Wide Tax News, 6 February 2025. <https://www.bdo.global/en-gb/insights/tax/world-wide-tax/indonesia-global-minimum-tax-under-pillar-two-implemented>.
- 92 United States Geological Survey. 2026. “Nickel.” In *Mineral Commodity Summaries 2026*. USGS, February 2026. <https://pubs.usgs.gov/periodicals/mcs2026/mcs2026-nickel.pdf>.
- 93 Investment Monitor. 2024. “BYD Inaugurates Its First EV Plant in Southeast Asia.” Invest-ment Monitor, July 2024. <https://www.investmentmonitor.ai/news/byd-inaugurates-its-first-ev-plant-in-southeast-asia/>.
- 94 The Nation Thailand. 2025. “EV Sales Soar to 18% Share as Thai Car Market Edges Up in 2025.” The Nation, 18 August 2025. <https://www.nationthailand.com/business/automobile/40054156>.
- 95 ASEAN Briefing. 2025. “How BOI EV 3.5 Shapes the Future of EV Battery Investments in Thailand.” ASEAN Briefing, 2 September 2025. <https://www.aseanbriefing.com/news/how-boi-ev-3-5-shapes-the-future-of-ev-battery-investments-in-thailand/>.
- 96 EY. 2025. “Thailand Enacts Top-up Tax Emergency Decree to Implement BEPS 2.0 Pillar Two.” EY Global Tax Alert, January 2025. <https://taxnews.ey.com/news/2025-0119-thailand-enacts-top-up-tax-emergency-decree-to-implement-beps-20-pillar-two>.
- 97 Automotive Logistics. 2026. “Thai Auto Groups Submit Proposal to Protect Local EV Pro-duction.” Automotive Logistics, 29 May 2026. <https://www.automotivelogistics.media/ev-and-battery/thai-automotive-associations-sign-joint-statement-to-government-proposing-eight-measures-to-prevent-ev-production-collapse/2666736>.
- 98 Energy-Storage.News. 2022. “Gotion Building Vietnam’s First LFP Gigafactory.” Energy-Storage.News, 21 November 2022. <https://www.energy-storage.news/gotion-building-vietnams-first-lpf-gigafactory/>.

99 International Energy Agency (IEA). 2025. *Global EV Outlook 2025: Electric Vehicle Charging*. IEA, 2025. <https://www.iea.org/reports/global-ev-outlook-2025/electric-vehicle-charging>.

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